

# ENGR489 Literature Notes

Oct 9, 2026 · @og\_bonglord

One note per source, keyed by its Zotero cite key. Each batch adds its rows to the index and its notes to the end of the Notes section.

## Index

| Cite key             | Short title                                                                | Batch | Supports claims                                                           | Verified |
|----------------------|----------------------------------------------------------------------------|-------|---------------------------------------------------------------------------|----------|
| wang2002improved     | An improved Helmholtz coil and analysis of its magnetic field homogeneity  | 1     | I-A (Helmholtz uniformity, flat-centre condition); IV-K / VI (supporting) | no       |
| sobko2024development | The development of a 3D magnetic field scanner using additive technologies | 1     | II-A; II-B; II-E; III-G (supporting)                                      | no       |
| senis_mms            | Senis Magnetic Field Mapper MMS-1A-RS / MMS-1X-RS (product page)           | 1     | I-A / II-A (cost, Cartesian mT-T mappers); III-B (supporting); IV-L       | no       |
| metrolab_hallinsight | Metrolab HallinSight 3D-Hall magnetic field camera (product page)          | 1     | II-A (sensor-array cameras); I-A / IV-L (cost)                            | no       |

| Cite key                | Short title                                                                               | Batch | Supports claims                                               | Verified |
|-------------------------|-------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------|----------|
| un2015transforming      | Transforming our world: the 2030 Agenda for Sustainable Development                       | 1     | I-D; III-K (supporting); IV-L (supporting)                    | no       |
| nasa_smallsats          | What are SmallSats and CubeSats?                                                          | 1     | Background only (I-A only if the cage is for CubeSat testing) | no       |
| nogiec2022software      | Software Architecture and Hardware Organization in Mu2e Solenoid Field Mapping System     | 2     | II-A; II-B; II-E; III-G (supporting)                          | no       |
| clavel1990device        | Device for the movement and positioning of an element in space (US Patent 4,976,582)      | 2     | II-C; II-B / III-J (supporting)                               | no       |
| williams2016delta       | The Delta Parallel Robot: Kinematics Solutions                                            | 2     | III-E; II-C; III-H (supporting)                               | no       |
| bentaleb2020improvement | On the improvement of calibration accuracy of parallel robots – modeling and optimization | 2     | II-D; II-E; IV-J                                              | no       |

| Cite key                   | Short title                                                                                                       | Batch | Supports claims                                               | Verified |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------|----------|
| tirathompson2008digital    | Digital Servo Calibration and Modeling                                                                            | 2     | II-D; III-F; III-H; IV-B (comparison)                         | no       |
| iso1998manipulating        | ISO 9283:1998 Manipulating industrial robots — Performance criteria and related test methods (ISO catalogue page) | 2     | II-F / IV-A (citation only)                                   | no       |
| pollak2023simulation       | Simulation Design and Measurement of Welding Robot Repeatability Utilizing the Contact Measurement Method         | 3     | II-F; IV-A (supporting); IV-D (comparison); IV-J (supporting) | no       |
| uscategui2023highprecision | High-Precision Magnetic Testbed Design and Simulation for LEO Small-Satellite Control Test                        | 3     | I-A; II-A; IV-K / VI; II-B (supporting); IV-G (context)       | no       |
| cervettini2021development  | Development and Experimentation of a CubeSat Magnetic Attitude Control System Testbed                             | 3     | I-A (only if CubeSat framing); IV-G (context)                 | no       |

| Cite key                      | Short title                                                                                                                                                      | Batch | Supports claims                                                                 | Verified |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------|----------|
| alken2021international        | International Geomagnetic Reference Field: the thirteenth generation                                                                                             | 3     | IV-G; I-A (supporting)                                                          | no       |
| noaa_magcalc                  | NCEI Geomagnetic Calculators — magnetic field calculator (saved input form, no result)                                                                           | 3     | IV-G (once rerun); Background only until then                                   | no       |
| tsekos2007magnetic            | Magnetic Resonance-Compatible Robotic and Mechatronics Systems for Image-Guided Interventions and Rehabilitation: A Review Study (PDF + Annual Reviews web page) | 4     | II-B; II-D / III-F; III-H (supporting); IV-E / IV-K (method); IV-D (comparison) | no       |
| nzminiaturebearings2026female | Female Rod End (SI6T/K) M6 x 1.0 Right Hand Thread Bearing (supplier product page)                                                                               | 4     | III-B; III-C; IV-K; IV-L (cost)                                                 | no       |

| Cite key                    | Short title                                                                                                                                         | Batch | Supports claims                                         | Verified |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------|----------|
| omelyanchik2023doityourself | A do-it-yourself approach for developing a magnetic field mapping setup using a 3D printer                                                          | 4     | I-A (cost); II-A; II-E; III-G; IV-L / I-D (SDG 4)       | no       |
| farooq2023decade            | A Decade of MRI Compatible Robots: Systematic Review                                                                                                | 4     | II-B; II-B / III-F; II-C (lead only); IV-D (comparison) | no       |
| yang2020analytical          | Analytical elastostatic stiffness modeling of parallel manipulators considering the compliance of the link and joint (PDF + ScienceDirect web page) | 5     | II-D; III-H (framework only); IV-J (supporting)         | no       |
| futaba2010s3306mg           | Futaba S3306MG servo datasheet                                                                                                                      | 5     | II-G; III-C; III-D (supporting)                         | no       |

## Notes

New notes go below, one ### heading per cite key, in batch order.

wang2002improved

**An improved Helmholtz coil and analysis of its magnetic field homogeneity.** Wang, She and Zhang, *Rev. Sci. Instrum.* 73(5), pp. 2175–2179, 2002. DOI 10.1063/1.1471352.

**Summary (3–4 lines):** The authors add a third, central coil of the same radius to a circular Helmholtz pair, wired in series. They choose the spacing and turns ratio so that the second and fourth derivatives of the axial field are zero at the centre:  $a/R = 0.7601$  and  $k = 0.5315$ , i.e. 59 central turns to 111 outer turns. They then compare the uniform regions of a Helmholtz pair, Maxwell's tricoil and the new coil numerically. The new coil gives the largest uniform region. The work is analytical only; the coil was not built.

### Key equations:

- Eq. (1), p. 2175: Taylor expansion of the axial field about the centre. It shows why  $B^{(2)}(0) = 0$  gives a flat field.  $z$  = axial distance from the centre (m).

$$B(z) = B(0) + \frac{1}{2}B^{(2)}(0)z^2 + \frac{1}{24}B^{(4)}(0)z^4 + \frac{1}{720}B^{(6)}(0)z^6 + \dots$$

- Eq. (2), p. 2175: on-axis field of a circular Helmholtz pair.  $N$  = turns per coil,  $I$  = current (A),  $R$  = coil radius = coil spacing (m),  $\mu_0$  = permeability of free space;  $B$  in T.

$$B(z) = \frac{1}{2}\mu_0 N I R^2 \left\{ \left[ R^2 + \left( z + \frac{R}{2} \right)^2 \right]^{-3/2} + \left[ R^2 + \left( z - \frac{R}{2} \right)^2 \right]^{-3/2} \right\}$$

- Eq. (3), p. 2175: field at the centre. This equals  $(4/5)^{3/2} \mu_0 N I / R \approx 0.7155 \mu_0 N I / R$ . That simplification is mine, not printed in the paper.

$$B(0) = \frac{16}{5\sqrt{5}} \frac{1}{2} \mu_0 N I R$$

- Eq. (4), p. 2175: relative deviation of the axial field from the centre value.  $\xi = z/R$  (dimensionless).

$$\epsilon = \frac{B(z) - B(0)}{B(0)} = \frac{5\sqrt{5}}{16} \left\{ \left[ 1 + \left( \frac{1}{2} + \xi \right)^2 \right]^{-3/2} + \left[ 1 + \left( \frac{1}{2} - \xi \right)^2 \right]^{-3/2} \right\} - 1$$

- Eqs. (5)–(9), p. 2176: off-axis  $B_r$  and  $B_z$  of the Helmholtz pair, using complete elliptic integrals  $K$  and  $E$ . Eq. (9) gives the off-axis deviation  $\epsilon_z$ , with  $\rho = r/R$ . Not transcribed; transcribe them only if you use them.
- Eqs. (14)–(20), p. 2177: the three-coil design, its derivative conditions and its  $\epsilon$ . Background only for this report.

### Figures / tables:

- Fig. 1, p. 2177: on-axis  $\epsilon$  against  $\xi$  for the Helmholtz, Maxwell and improved coils. It shows how fast the Helmholtz field falls off with axial distance.
- Figs. 2–3, p. 2178: off-axis  $\epsilon_z$  against  $\rho$  at  $\epsilon = 1\%$  and  $\epsilon = 0.1\%$ .
- Table I, p. 2178: uniform-region sizes. This is useful as a single citable number for how uniform a Helmholtz coil is.
  - Within 1 %:  $|z| \leq 0.3137R$  and  $r \leq 0.2744R$ .
  - Within 0.1 %:  $|z| \leq 0.1731R$  and  $r \leq 0.1491R$ .
- Text on p. 2178: treating the uniform regions as cylinders, the improved coil's volume is 2.20× the Helmholtz pair's at 1 % and 4.19× at 0.1 %. Maxwell's tricoil is 1.27× and 2.24×.

### Assumptions / limitations:

- Circular, thin coils (filament model), with no winding cross-section, tolerances or nearby materials.
- Purely numerical; there are no measurements.
- Wilson's cage uses **square** coils, so these formulas and region sizes don't apply to it directly.

### Use in my report:

- Section: I-A Problem statement. Claim: "A Helmholtz pair's field is uniform only near its centre. For circular coils it stays within 1 % of the centre value for  $|z| \leq 0.31R$  and  $r \leq 0.27R$ , so the usable volume must be mapped."
- Section: I-A. Claim: "The field is flat at the centre because the coil spacing makes the second derivative of the axial field zero." This rests on Eq. (1) with  $B^{(2)}(0) = 0$ .
- Section: IV-K / VI, supporting only. This is the theory the future coils-on map would be compared with, once a square-coil equivalent is found.

### Unclear / check:

- Eq. (15) is printed as " $B^2(0)$ ". From context it means the second derivative  $B^{(2)}(0)$ .
- The paper says "magnetic field intensity" but writes  $B$  with  $\mu_0$ , so it means flux density in T.
- Gap: you need a square-coil Helmholtz source for Wilson's cage.

sobko2024development

**The development of a 3D magnetic field scanner using additive technologies.** Sobko, Yudanov, Panina and Rodionova, *Hardware* 2(4), pp. 279–291, 2024. DOI 10.3390/hardware2040014. Open access (CC BY).

**Summary (3–4 lines):** A low-cost Cartesian scanner moves an HMC5883 three-axis magnetometer (GY-273 module) through a  $100 \times 100 \times 100$  mm volume.

- **Build:** an aluminium-profile frame with 3D-printed ABS brackets, lead-screw guides and gears. Small stepper motors sit outside the scan area and are switched off during each reading.
- **Electronics and software:** a PIC16F877A drives L298N stepper drivers and streams data to Excel through the PLX-DAQ macro. The electronics cost under US\$35 (p. 289).
- **Validation:** tested on a PCB flat coil (17 turns, 0.6 A) against a Biot–Savart model.  $H_z$  matched closely;  $H_x$  showed an offset that the authors attribute to Earth's field.

**Key equations** (all on p. 287;  $H$  in A/m,  $I$  in A, lengths in m):

- Eq. (1): Biot–Savart law for a current element  $I d\mathbf{l}$  at distance  $r$ .

$$d\mathbf{H} = \frac{1}{4\pi} \frac{I d\mathbf{l} \times \mathbf{r}}{r^3}$$

- Eq. (2):  $H_x$  in the x-z plane for  $N$  coaxial circular loops of radius  $a_i$ .  $\varphi$  = angle around the loop.

$$H_x(x, z) = \frac{I}{4\pi} \sum_{i=1}^N \int_0^{2\pi} \frac{a_i \sin\varphi}{\left(a_i^2 + x^2 + z^2 - 2 a_i x \sin\varphi\right)^{3/2}} d\varphi$$

- Eq. (3):  $H_z$  for the same loops.

$$H_z(x, z) = \frac{I}{4\pi} \sum_{i=1}^N \int_0^{2\pi} \frac{a_i \left(a_i - x \sin\varphi\right)}{\left(a_i^2 + x^2 + z^2 - 2 a_i x \sin\varphi\right)^{3/2}} d\varphi$$

### Figures / tables:

- Fig. 1, p. 281: block diagram showing the PC, scanner, power supply and coil under test.
- Figs. 2–3, pp. 281–282: OpenSCAD model and layout. The motors are at the frame edge, outside the scan area.
- Fig. 6, p. 284: photo of the built scanner.
- Table 1, pp. 285–286: specs.
  - Full scale 1 mT; noise 10  $\mu$ T; non-linearity 0.1 %; hysteresis 25 ppm.
  - Sampling 0.75–75 Hz; spatial resolution 0.5–1 mm.
- Text on p. 284: scan times. A 2 mm-step plane ( $47 \times 50 = 2350$  points) takes about 20 min; 50 planes take nearly 17 h.

- Figs. 9–13, pp. 287–289: modelled and measured  $H_x$ ,  $H_y$  and  $H_z$  maps of the flat coil.
- Fig. 14, p. 289: measured vs modelled  $H_z(z, x = 28 \text{ mm})$  and  $H_x(z, x = 39 \text{ mm})$ .

### Assumptions / limitations:

- Not fully non-magnetic. The frame uses aluminium beams and "metal beams", and the steppers are small but still contain magnets.
- Positioning accuracy and repeatability are never measured; only step resolution is given. Homing with optical sensors before each measurement corrects accumulated error.
- Validation is qualitative (a curve overlay), with no error statistics.
- Sensor noise of  $10 \mu\text{T}$  is coarse for Earth-field-level mapping.

### Use in my report:

- Section: II-A Magnetic field mapping systems. Claim: "Low-cost DIY mappers build the frame from 3D-printed parts and cheap sensors, cutting electronics cost to tens of dollars, but they use Cartesian gantries with metal frames."
- Section: II-B Positioning in magnetic environments. Claim: "A common tactic is to keep the motors outside the scan volume and switch them off while sampling."
- Section: II-E How this project differs. Claim: "Prior low-cost scanners report step resolution, not measured position accuracy or repeatability."
- Section: III-G Scan software, supporting: switching the drives off during readings to avoid interference.

### Unclear / check:

- The paper calls the HMC5883 a Hall sensor (p. 280) but also says its element is anisotropic magnetoresistive (p. 282). The datasheet says AMR, so don't repeat "Hall".
- Pages 279–280 give geomagnetic fields as about  $10 \mu\text{T}$ . Earth's field is roughly  $25\text{--}65 \mu\text{T}$ , so don't cite this figure.
- Gap lead: their ref. [21], Omelyanchik et al., *Meas. Sci. Technol.* 34, 107001 (2023). It's a DIY mapper built on a 3D printer, and its ferromagnetic parts limited it to about 1 G. Good for II-A/II-B; add it to Zotero.

senis\_mms

### Senis Magnetic Field Mapper MMS-1A-RS / MMS-1X-RS (product page). GMW

Associates (US distributor for Senis), product web page, accessed 9 Oct 2026. A web page, so it has no page or equation numbers.

**Summary (3–4 lines):** A commercial three-axis Hall-probe mapper on a Cartesian moving platform, with a rotating stage. It measures all three field components at virtually the same point and shows 2D/3D colour maps. It's aimed at permanent magnets and electromagnets in QA and lab work. Listed prices are US\$143,180 for the MMS-1A-RS and US\$172,390 for the larger MMS-1X-RS.

**Key equations:** none.

**Figures / tables** (the page's spec table and price list):

| Item                                   | MMS-1A-RS       | MMS-1X-RS       |
|----------------------------------------|-----------------|-----------------|
| Scan volume (mm)                       | 125 × 125 × 150 | 570 × 570 × 150 |
| Price (USD, US/Canada/Mexico delivery) | 143,180         | 172,390         |

Specs shared by both models:

- **Motion:** 1  $\mu\text{m}$  linear and 0.022° rotary resolution, at up to 50 mm/s.
- **Field:** accuracy typically 0.1 % of full range. Ranges are  $\pm 100$  mT,  $\pm 500$  mT or  $\pm 2000$  mT, with DC–25 kHz bandwidth on all three axes.
- **AMR probe option (US\$18,860):** 0.5 mT range, resolution better than 100 nT, accuracy better than 200 nT.
- **Safety:** a cabinet with an e-stop (US\$6,600) is required for CE marking. A touch sensor on the probe also acts as an e-stop.

**Assumptions / limitations:**

- These are vendor claims, not independent measurements. Prices exclude US tariffs, are quoted FCA San Carlos, and may change.
- The system is built for mT-to-T fields. On the  $\pm 100$  mT range, 0.1 % of full range is about 100  $\mu\text{T}$  (my arithmetic). That is larger than a typical Helmholtz-cage field, so only the AMR option suits  $\mu\text{T}$  work.

**Use in my report:**

- Section: I-A Problem statement / II-A. Claim: "Commercial 3D field mappers cost well over €25,000; the Senis MMS-1A-RS lists at US\$143,180."
- Section: II-A. Claim: "Commercial mappers use Cartesian stages with  $\mu\text{m}$  resolution and Hall probes ranged for mT–T fields."

- Section: III-B, supporting: commercial practice uses an e-stop cabinet and a probe touch sensor.
- Section: IV-L Sustainability impact: cost comparison for lab access.

#### Unclear / check:

- The page title says "MMS-1-R", but the model names are MMS-1A-RS and MMS-1X-RS. Use the model names.
- For IEEE, cite it as a web page with the access date. The Senis datasheet PDF linked on the page is a more stable source.
- The page doesn't say whether the stage's motors sit near the probe.

#### metrolab\_hallinsight

**Metrolab HallinSight 3D-Hall magnetic field camera (product page).** GMW Associates (US distributor for Metrolab), product web page, accessed 9 Oct 2026. The technology was developed by Fraunhofer IIS. A web page, so it has no page or equation numbers.

**Summary (3–4 lines):** A "field camera": a planar array of three-axis Hall sensors that images all three field components over a line or plane in real time, with no moving stage. Listed applications are magnet QA, AC-field visualisation and motor QC.

**Key equations:** none.

**Figures / tables** (the page's text and price list):

| Model     | Sensors | Active area | Frame rate | Price (USD) |
|-----------|---------|-------------|------------|-------------|
| 32×2-4MR  | 64      | 80 × 2.5 mm | 250 Hz     | 17,950      |
| 16×16-4MR | 256     | 40 × 40 mm  | 100 Hz     | 28,490      |
| 32×32-4MR | 1024    | 80 × 80 mm  | 25 Hz      | 56,980      |

Other specs: range from "several  $\mu\text{T}$ " up to a few T; 4  $\mu\text{T}$  resolution; USB with an ASCII protocol; LabVIEW software; built-in temperature sensor.

#### Assumptions / limitations:

- These are vendor claims. Prices are for US/Canada/Mexico delivery, exclude tariffs, and may change.
- It covers at most 80 × 80 mm in one plane, so mapping a 3D volume still means moving the camera.

- 4  $\mu\text{T}$  resolution is coarse next to a  $\mu\text{T}$ -level Helmholtz map.

### Use in my report:

- Section: II-A Magnetic field mapping systems. Claim: "Sensor-array cameras image a plane at once with no moving parts, but cover at most  $80 \times 80$  mm and resolve about 4  $\mu\text{T}$ ."
- Section: I-A / IV-L: cost of commercial systems (US\$17,950–56,980).

### Unclear / check:

- The  $32 \times 2$  line camera at US\$17,950 is below €25,000 (see Flags).
- The page also says the sensors resolve the field "at a rate of up to 1 kHz", which doesn't match the 25–250 Hz array rates. Check the HallinSight brochure or key-specs PDF linked on the page, and cite that instead.

## un2015transforming

**Transforming our world: the 2030 Agenda for Sustainable Development.** UN General Assembly, resolution A/RES/70/1, adopted 25 Sep 2015. Your copy is the UN DESA web page (accessed 9 Oct 2026), which has paragraph and target numbers but no page numbers.

**Summary (3–4 lines):** The resolution sets 17 Sustainable Development Goals and 169 targets for 2030. It frames them as integrated and indivisible across the economic, social and environmental dimensions (Preamble; paras. 2 and 5). Progress is tracked through a global indicator framework (para. 75).

**Key equations:** none.

**Figures / tables** — the targets relevant to this report, paraphrased:

| Target | What it asks                                                                               | Where it fits                                                            |
|--------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 4.3    | Equal access to affordable, quality technical and tertiary education, including university | Low-cost lab equipment widens access to hands-on teaching                |
| 4.4    | More youth and adults with relevant technical and vocational skills                        | Student-built instrumentation as skills development                      |
| 9.5    | Enhance scientific research and upgrade technological capability, including more R&D       | An open, cheap mapper enables research the lab couldn't otherwise afford |

| Target | What it asks                                                   | Where it fits                                                                      |
|--------|----------------------------------------------------------------|------------------------------------------------------------------------------------|
| 9.b    | Support domestic technology development and research           | Optional: locally built rather than imported equipment                             |
| 12.2   | Sustainable management and efficient use of natural resources  | Material choice: PLA, carbon fibre, small part count                               |
| 12.4   | Sound management of chemicals and wastes over their life cycle | Battery choice and disposal                                                        |
| 12.5   | Cut waste through prevention, reduction, recycling and reuse   | Reused GUI and open-source code; PLA end of life; rechargeable vs disposable cells |

### Assumptions / limitations:

- The targets are global and policy-level. Linking a single lab instrument to them is an argument you make, not something the Agenda says.
- Target 12.4 had a 2020 deadline. If you use it for batteries, say so, or lean on 12.5 instead.

### Use in my report:

- Section: I-D Sustainability and the SDGs. Claim: "The UN 2030 Agenda sets 17 SDGs and 169 targets; this project addresses targets 4.3/4.4, 9.5 and 12.2/12.5."
- Section: III-K, supporting the material and battery choices (12.2, 12.4, 12.5).
- Section: IV-L, supporting the access and cost argument (4.3, 9.5).

### Unclear / check:

- Your outline says "SDGs 4, 9.5, 12". Decide which exact targets under 4 and 12 you'll name, and keep them the same in I-D, III-K and IV-L.
- For IEEE, cite the resolution itself (UN General Assembly, A/RES/70/1, 2015), not the DESA page. Check that your Zotero entry is in that form.

### nasa\_smallstats

**What are SmallSats and CubeSats?** NASA Small Spacecraft Systems Virtual Institute (S3VI), web page, last updated 21 Sep 2026, accessed 9 Oct 2026. No page numbers.

**Summary (3–4 lines):** The page defines small spacecraft as under 180 kg, in five classes:

- mini: 100–180 kg
- micro: 10–100 kg

- nano: 1–10 kg
- pico: 0.01–1 kg
- femto: 0.001–0.01 kg

CubeSats are a standard nanosatellite form factor built from  $10 \times 10 \times 10$  cm "1U" units, scalable from 1.5U to 12U. Cal Poly and Stanford introduced them in 1999 as a platform for education and exploration. They are now a cheap platform for science and technology demonstrations.

**Key equations:** none.

**Figures / tables:** none useful.

**Assumptions / limitations:**

- Says nothing about magnetic testing, Helmholtz cages, magnetometers or attitude control.
- Overview level only, with mission history specific to NASA Ames.

**Use in my report:**

- Background only, unless your report says Wilson's cage is for CubeSat testing. In that case: Section: I-A. Claim: "CubeSats are standardised nanosatellites built from 10 cm cubic units and are widely used in universities."

**Unclear / check:**

- This page can't support a claim that Helmholtz cages are used to test CubeSat magnetometers and magnetorquers. If you want to make that claim, you need a source that says so, such as a paper on a CubeSat attitude-control test facility.

nogiec2022software

**Software Architecture and Hardware Organization in Mu2e Solenoid Field Mapping System.** J. M. Nogiec, P. Akella, S. Feher, J. Grudzinski, T. Strauss, R. L. Talaga and L. Xia, *IEEE Trans. Appl. Supercond.*, vol. 32, no. 6, Sep. 2022, Art. no. 9000704 (4 pp.). DOI 10.1109/TASC.2022.3155474.

**Summary (3–4 lines):** Fermilab and Argonne describe the automated system that maps the Mu2e detector solenoid (10.9 m long; field graded from 2 T to 1 T, then a uniform 1 T). A self-propelled mapper runs on rails inside the bore and rotates two "propeller" arms carrying 3D Hall probes and an NMR probe. A Leica AT403 laser tracker measures retroreflectors on the arms, so each probe's position and orientation are known. The

software is a component-based publish-subscribe framework (EMMA), driven by Python scripts, with automatic quality checks. It's a design paper: no measured maps or achieved accuracies are reported.

**Key equations:** none (no numbered equations). Numerical requirements, pp. 1–2:

- Target-region gradient  $-0.25 \pm 0.05$  T/m. Allowed non-uniformity  $\Delta B/B$ : 0.01 % (spectrometer), 0.05 % (calorimeter).
- Accuracy 0.01 % (1 G in 1 T). Field direction known to better than 0.1 mrad.
- For comparison: CMS mapping precision 0.07 %; ATLAS probes 0.01 % up to 1.4 T and 0.05 % up to 2.5 T, alignment  $\pm 2$  mrad.

**Hardware** (pp. 2–4):

- **Mapper:** mostly aluminium. The shaft between the propellers is carbon fibre, the reduction gears stainless steel, the bearings plastic and the screws brass.
- **Drive:** Shinsei USR60-E3NT ultrasonic motors and a D6060E driver, both on the mapper and chosen to be non-magnetic. Galil DMC-4040 motion controller.
- **Hall probes:** BsCAN3 3D probes (CERN/NIKHEF), each with three Siemens KSY44 Hall sensors and a Cirrus Logic CS5524 24-bit ADC. Read over CAN in under a second.
- **NMR:** Metrolab PT2026,  $\pm 5$  ppm.
- **Laser tracker:** Leica AT403. It works in fields up to 200 G, with typical distance accuracy  $\pm 0.005$  mm. Required tracker accuracy for the large propeller is 0.07 mm (p. 4).

**Figures / tables:**

- Fig. 1, p. 1: the detector solenoid and its axial field (2 T falling to 1 T, then flat).
- Fig. 2, p. 2: photo of mapper integration testing at Argonne.
- Fig. 3, p. 2: CAD of the mapper, with large and small propellers, Hall probes and tracker targets.
- Fig. 4, p. 3: software components, grouped as framework, UI, DAQ/control, data processing and saving.
- No tables.

**Assumptions / limitations:**

- A design and architecture paper, with no measured accuracy or map results.
- Far bigger scale and budget than this project (1 T superconducting solenoid, laser tracker).

- Set-up is manual: whenever the tracker moves, an operator points it at each target (p. 4).
- Finding the targets during a scan depends on a model of the mapper predicting where they will be.

### Use in my report:

- Section: II-A Magnetic field mapping systems. Claim: "Accelerator magnets are mapped with purpose-built mappers carrying 3D Hall and NMR probes, at about 0.01 % accuracy (Mu2e, ATLAS, CMS)."
- Section: II-B Positioning in magnetic environments. Claim: "Mappers that work inside the field use non-magnetic construction: aluminium, carbon fibre, plastic bearings, brass screws and ultrasonic motors."
- Section: II-E How this project differs. Claim: "Large mappers locate their probes with external metrology (a laser tracker,  $\pm 0.005$  mm); this project checks position in situ with the magnetometer."
- Section: III-G Scan software, supporting: fully scripted scans with per-step quality checks and logging.

### Unclear / check:

- Arm spans are given as "64.96" and "30" (large propeller) and "9.45" and "8.5" (small), with no units (p. 2). They're probably inches; don't cite them.
- "0.01 % (1 G in 1 T)" is self-consistent ( $1 \text{ G} = 10^{-4} \text{ T}$ ).
- Gap leads: ref. [3] Feher et al. 2018, Mu2e field mapping system design, *IEEE TAS* 28(3) 9001305; ref. [7] Orozco et al. 2018, Hall probe calibration for Mu2e, *IEEE TAS* 28(3) 9500504; ref. [9] Lippitsch et al., Leica laser tracker operation in magnetic fields (IWAA2018); refs. [4] Klyukhin 2008 (CMS) and [5] Alexa 2008 (ATLAS).

clavel1990device

**Device for the movement and positioning of an element in space.** R. Clavel, U.S. Patent 4,976,582, granted 11 Dec. 1990. Assignee Sogeva S.A. Appl. no. 403,987, filed 6 Sep. 1989 (continuation of Ser. no. 96,113, filed 13 Aug. 1987). Swiss priority CH 5348/85, 16 Dec. 1985. Certificate of correction issued 4 Aug. 1992. 11 pp. (4 drawing sheets, 25 claims).

**Summary (3–4 lines):** This is the original delta robot patent. Three actuators fixed to a base each turn a rigid control arm. Each arm connects to a small movable platform through a pair of parallel bars with cardan or ball joints. The parallelograms keep the platform parallel to the base whatever the arms do, so it only translates (3 DOF). An

optional fourth DOF rotates a tool through a telescopic arm driven by a base-mounted motor. Variants use single bars with cardan joints (Figs. 3–4) or linear slides in place of the rotary arms (Fig. 5).

**Key equations:** none.

**Figures / tables:**

- Fig. 1, p. 2 (sheet 1): the main embodiment. It shows base 1, rotary actuators 13, arms 4, parallel bars 5a/5b, platform 8, telescopic arm 14, motor 11 and computer 12.
- Fig. 2, p. 3 (sheet 2): a version with ball-and-socket joints and motor 11 on the platform.
- Figs. 3–4, p. 4 (sheet 3): a single-bar version with cardan joints 36 and 37.
- Fig. 5, p. 5 (sheet 4): a linear-actuator version, with members 24 on straight guides.

**Key passages:**

- Col. 1, lines 15–40 (p. 6): serial robots carry a motor on each axis, so the moving mass is large even for a small load.
- Col. 3, lines 1–11 (p. 7): the advantages claimed are a large reachable volume, minimal moving mass and very high rates. The three parallel arms add stiffness, which gives better repeatability than most industrial robots.
- Col. 4, lines 37–44 (p. 7): listed uses include assembly, packing, and finding a position in space, e.g. measuring a displacement or dimensional checking.
- Claim 1, cols. 4–5 (pp. 7–8): the main independent claim.

**Assumptions / limitations:**

- A patent: no kinematics, performance or accuracy data.
- The repeatability advantage is asserted, not measured.

**Use in my report:**

- Section: II-C Delta robots. Claim: "Clavel's delta robot (Swiss priority 1985, US patent 1990) uses three base-mounted rotary actuators and parallelogram forearms to hold the platform at constant orientation, giving 3-DOF translation with low moving mass."
- Section: II-B / III-J, supporting. Claim: "Because every actuator is fixed to the base, a delta can keep its motors far from the end effector." This backs putting the servos more than 600 mm above the probe.
- Section: III-J Design alternatives. Claim: "Parallel arms are stiffer than a serial chain and avoid carrying motors on the moving links" (cols. 1 and 3).
- Optional, II-C: Clavel already listed position measurement as a use.

**Unclear / check:**

- The certificate of correction (pp. 10–11) fixes typos in the printed text, e.g. "oars" should read "bars" (col. 3, line 55). Use the corrected wording when paraphrasing.
- The Fig. 1 design is 4-DOF (translation plus tool rotation). Your robot is the 3-DOF translation-only form.
- IEEE patent format: R. Clavel, "Device for the movement and positioning of an element in space," U.S. Patent 4 976 582, Dec. 11, 1990.
- Gap lead: Clavel's 1991 EPFL PhD thesis, "Conception d'un robot parallèle rapide à 4 degrés de liberté" (cited by Williams, fn. 1).

williams2016delta

**The Delta Parallel Robot: Kinematics Solutions.** R. L. Williams II, Mechanical Engineering, Ohio University. Internet publication, 2016, 46 pp. URL: [www.ohio.edu/people/williar4/html/pdf/DeltaKin.pdf](http://www.ohio.edu/people/williar4/html/pdf/DeltaKin.pdf) (accessed 9 Oct. 2026). No DOI.

**Summary (3–4 lines):** A tutorial that derives the kinematics of the 3-DOF translational delta robot, for revolute inputs (Clavel's type) and prismatic inputs (linear delta 3D printers). Inverse kinematics is solved one leg at a time as a circle–sphere intersection. This reduces to  $E \cos \theta + F \sin \theta + G = 0$ , which is solved with the tangent half-angle substitution. Forward kinematics is the intersection of three spheres centred on the knees, shifted by the platform offsets. The paper also gives velocity equations and worked examples with ABB FlexPicker IRB 360 dimensions. I re-ran the revolute and prismatic snapshot examples, and they reproduce.

**Key equations** (revolute input; the body equations are unnumbered, so pages only):

- p. 6: equilateral base and platform geometry.  $s_B$  and  $s_P$  are the base and platform sides (m);  $w$  is the inradius and  $u$  the circumradius.

$$w_B = \frac{\sqrt{3}}{6}s_B, \quad u_B = \frac{\sqrt{3}}{3}s_B, \quad w_P = \frac{\sqrt{3}}{6}s_P, \quad u_P = \frac{\sqrt{3}}{3}s_P$$

- p. 6: hip points  $B_i$  in the base frame  $\{B\}$ , and platform joint points  $P_i$  in the platform frame  $\{P\}$ .

```

{{}^{\{B\}}\mathbf{B}}_1=\begin{Bmatrix}0\\-w_B\\0\end{Bmatrix},\;
{{}^{\{B\}}\mathbf{B}}_2=\begin{Bmatrix}\tfrac{\sqrt{3}}{2}w_B\\\tfrac{1}{2}w_B\\0\end{Bmatrix},\;{{}^{\{B\}}\mathbf{B}}_3=\begin{Bmatrix}-\tfrac{\sqrt{3}}{2}w_B\\\tfrac{1}{2}w_B\\0\end{Bmatrix},\quad
{{}^{\{P\}}\mathbf{P}}_1=\begin{Bmatrix}0\\-u_P\\0\end{Bmatrix},\;{{}^{\{P\}}\mathbf{P}}_2=\begin{Bmatrix}\tfrac{s_P}{2}\\w_P\\0\end{Bmatrix},\;{{}^{\{P\}}\mathbf{P}}_3=\begin{Bmatrix}-\tfrac{s_P}{2}\\w_P\\0\end{Bmatrix}

```

- p. 7: Kutzbach mobility.  $M = \text{DOF}$ ;  $N = \text{links including ground}$ ;  $J_1, J_2$  and  $J_3 = \text{numbers of 1-, 2- and 3-DOF joints}$ . Counting every link gives  $M = -9$  (overconstrained). Removing one parallelogram bar per leg ( $N = 14, J_1 = 15$ ) gives  $M = 3$ .

$$M = 6(N-1) - 5J_1 - 4J_2 - 3J_3$$

- p. 10: vector loop for each leg  $i$ .  $L_i = \text{upper-arm vector (length } L, m)$ ;  $l_i = \text{forearm vector (length } l, m)$ ; the platform rotation is the identity.

```

{{}^{\{B\}}\mathbf{B}}_i + {{}^{\{B\}}\mathbf{L}}_i + {{}^{\{B\}}\mathbf{l}}_i =
{{}^{\{B\}}\mathbf{P}}_P + {{}^{\{B\}}_{\{P\}}\mathbf{R}}, {{}^{\{P\}}\mathbf{P}}_i, \quad
{{}^{\{B\}}_{\{P\}}\mathbf{R}} = \mathbf{I}_3, \quad i = 1, 2, 3

```

- p. 10: upper-arm vectors.  $\theta_i = \text{actuated joint angle (rad)}$ , zero with the arm horizontal, measured by the right-hand rule about the axes shown on p. 3.

```

{{}^{\{B\}}\mathbf{L}}_1=\begin{Bmatrix}0\\-L\cos\theta_1\\L\sin\theta_1\end{Bmatrix},\;{{}^{\{B\}}\mathbf{L}}_2=\begin{Bmatrix}\tfrac{\sqrt{3}}{2}L\cos\theta_2\\\tfrac{1}{2}L\cos\theta_2\\-L\sin\theta_2\end{Bmatrix},\;
{{}^{\{B\}}\mathbf{L}}_3=\begin{Bmatrix}-\tfrac{\sqrt{3}}{2}L\cos\theta_3\\\tfrac{1}{2}L\cos\theta_3\\-L\sin\theta_3\end{Bmatrix}

```

- p. 11: constants, and the three constraint equations from setting each forearm length to  $l$ .  $(x, y, z) = \text{platform centre in } \{B\} \text{ (m)}$ .

$$a = w_B - u_P, \quad b = \frac{s_P}{2} - \frac{\sqrt{3}}{2}w_B, \quad c = w_P - \frac{1}{2}w_B$$

```

\begin{aligned}
&2L(y+a)\cos\theta_1 + 2zL\sin\theta_1 + x^2+y^2+z^2+a^2+L^2+2ya-1^2 = 0\\
&-L\left(\sqrt{3}(x+b)+y+c\right)\cos\theta_2 + 2zL\sin\theta_2 + \\
&x^2+y^2+z^2+b^2+c^2+L^2+2xb+2yc-1^2 = 0\\
&L\left(\sqrt{3}(x-b)-y-c\right)\cos\theta_3 + 2zL\sin\theta_3 + \\
&x^2+y^2+z^2+b^2+c^2+L^2-2xb+2yc-1^2 = 0
\end{aligned}

```

- pp. 11-12: inverse kinematics. Each leg gives one EFG equation; the half-angle substitution turns it into a quadratic in  $t_i$ . The two roots are knee-in and knee-out (8 combinations); the knees-out set is chosen.

$$E_i \cos\theta_i + F_i \sin\theta_i + G_i = 0$$

```

\begin{aligned}
E_1 &= 2L(y+a), & F_1 &= 2zL, & G_1 &= x^2+y^2+z^2+a^2+L^2+2ya-1^2 \\
E_2 &= -L\left(\sqrt{3}(x+b)+y+c\right), & F_2 &= 2zL, & G_2 &= \\
&x^2+y^2+z^2+b^2+c^2+L^2+2(xb+yc)-1^2 \\
E_3 &= L\left(\sqrt{3}(x-b)-y-c\right), & F_3 &= 2zL, & G_3 &= \\
&x^2+y^2+z^2+b^2+c^2+L^2+2(-xb+yc)-1^2
\end{aligned}

```

$$t_i = \tan\frac{\theta_i}{2}, \quad t_{i\{1,2\}} = \frac{-F_i \pm \sqrt{E_i^2 + F_i^2 - G_i^2}}{G_i - E_i}, \quad \theta_i = 2\tan^{-1}(t_i)$$

- p. 13: forward kinematics. The platform centre is where three spheres of radius  $l$  meet. Their centres are the knee points  $A_i$  shifted back by the platform vertices.

$$\mathbf{A}_i = \mathbf{B}_i + \mathbf{L}_i, \quad \mathbf{A}_{iv} = \mathbf{A}_i - \mathbf{P}_i$$

- p. 15: velocity kinematics, from differentiating the constraints.  $\dot{X} = (\dot{x}, \dot{y}, \dot{z})$  in m/s;  $\dot{\Theta} =$  joint rates in rad/s.

$$[\mathbf{A}]\{\dot{\mathbf{X}}\} = [\mathbf{B}]\{\dot{\boldsymbol{\Theta}}\}$$

$$\mathbf{A} = \begin{bmatrix} x & y+a+L\cos\theta_1 & z+L\sin\theta_1 \\ 2(x+b)-\sqrt{3}L\cos\theta_2 & 2(y+c)-L\cos\theta_2 & 2(z+L\sin\theta_2) \\ 2(x-b)+\sqrt{3}L\cos\theta_3 & 2(y+c)-L\cos\theta_3 & 2(z+L\sin\theta_3) \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & 0 \\ 0 & 0 & b_{33} \end{bmatrix}$$

```

\begin{aligned}
b_{11} &= L\left[(y+a)\sin\theta_1 - z\cos\theta_1\right] \\
b_{22} &= -L\left[\left(\sqrt{3}(x+b)+y+c\right)\sin\theta_2 + 2z\cos\theta_2\right] \\
b_{33} &= L\left[\left(\sqrt{3}(x-b)-y-c\right)\sin\theta_3 - 2z\cos\theta_3\right]
\end{aligned}

```

The paper doesn't write out the revolute Jacobian. It follows as  $J = A^{-1}B$ , the form the paper uses for the prismatic case (p. 35).

- Eqs. (B.9)–(B.12), pp. 45–46: three-sphere intersection when all centres share one height  $z_n$ . This applies whenever the three joint angles are equal, where the general algorithm of Appendix A divides by zero (p. 14). Centres  $(x_j, y_j, z_n)$  and radii  $r_j$  in m; a–f here are local symbols, not the leg constants.

```

\begin{bmatrix}a & b \\ d & e\end{bmatrix}\begin{Bmatrix}x \\ y\end{Bmatrix} = \begin{Bmatrix}c \\ f\end{Bmatrix}, \quad
\begin{aligned}
a &= 2(x_3-x_1), & b &= 2(y_3-y_1), & c &= r_1^2-r_3^2-x_1^2-y_1^2+x_3^2+y_3^2 \\
d &= 2(x_3-x_2), & e &= 2(y_3-y_2), & f &= r_2^2-r_3^2-x_2^2-y_2^2+x_3^2+y_3^2
\end{aligned}

```

$$x = \frac{ce-bf}{ae-bd}, \quad y = \frac{af-cd}{ae-bd}$$

$$z^2 + Bz + C = 0, \quad B = -2z_n, \quad C = z_n^2 - r_1^2 + (x-x_1)^2 + (y-y_1)^2, \quad z = \frac{-B \pm \sqrt{B^2-4C}}{2}$$

Take the root below the base (negative  $z$ ).

- Eqs. (A.1)–(A.21), pp. 41–44: the general three-sphere algorithm and its singularities. These include  $a_{13} = 0$  or  $a_{23} = 0$  when two centres share a height. Background only; not transcribed.
- Prismatic input, pp. 27–35: inverse kinematics  $L_i = -z \pm \sqrt{z^2 - C_i}$  (p. 28), a closed-form forward solution (pp. 33–34) and velocities (p. 35). Background only; not transcribed.

### Figures / tables:

- p. 1: Clavel's patent Fig. 1 next to a photo of an ABB FlexPicker.

- p. 4: kinematic diagram, with hips  $B_i$ , knees  $A_i$ , ankles  $P_i$ , upper arm  $L$ , forearm  $l$  and parallelogram width  $h$ .
- p. 5: base and platform triangle details.
- Table, p. 6: FlexPicker IRB 360-1/1600 dimensions, scaled from a figure:  $s_B$  567,  $s_P$  76,  $L$  524,  $l$  1244 and  $h$  131 mm.
- p. 9: practical delta robots, including a hobby servo "Sketchy" drawing delta.
- p. 13: FPK diagram showing the three virtual spheres.
- pp. 16–21: IPK and FPK examples and trajectory plots.  $(0, 0, -0.9)$  m gives  $\theta = -20.5^\circ$  for all three joints.  $(0.3, 0.5, -1.1)$  m gives  $(47.5, -11.6, 21.4)^\circ$ .  $\theta = 0$  gives  $z = -1.065$  m, and  $(10, 20, 30)^\circ$  gives  $(0.108, -0.180, -1.244)$  m.
- Table, p. 25: Delta Maker 3D printer parameters (prismatic case).

### Assumptions / limitations:

- Ideal rigid links and joints: no offsets, compliance, backlash or manufacturing error.
- Base and platform are equilateral and the legs identical. The author says the method can be adapted to the general case.
- A tutorial, not peer-reviewed.
- No singularity analysis of the revolute velocity equations.

### Use in my report:

- Section: III-E Kinematic model. Claim: "Inverse kinematics is solved per leg as a circle-sphere intersection. This gives  $E \cos \theta + F \sin \theta + G = 0$ , which is solved with the tangent half-angle substitution, taking the knees-out root."
- Section: III-E. Claim: "Forward kinematics is the intersection of three spheres of radius  $l$  centred on the knee points shifted by the platform vertices. The equal-height case needs its own algorithm."
- Section: III-E / III-H, supporting. Claim: "Differentiating the constraints gives  $AX = B\dot{\theta}$ , so the Jacobian is  $J = A^{-1}B$ ." The compliance correction builds on this; the  $J^T$  step needs another source.
- Section: II-C Delta robots. Claim: "Delta kinematics have closed-form inverse and forward solutions."

### Unclear / check:

- The date is inconsistent. The title page says October 2016, the "for referencing" line says January 2016, and the PDF was created on 15 Jan. 2016. Cite it as the author asks (Jan. 2016).

- p. 6 says the  $P_i$  are constant "in the base frame  $\{P\}$ ". It should say the platform frame. A similar slip on p. 24 puts superscript P on the base points  $B_i$ .
- p. 37 (prismatic trajectory) says  $z$  is centred on  $-1$  m with  $0.05$  m amplitude, but the plots on p. 38 are centred on  $-0.5$  m. The text is wrong.
- p. 19 gives  $z = -1.065$  m for  $\theta = 0$ ; my recomputation gives  $-1.0645$  m (rounding).
- Check his  $\theta$  zero and sign against your firmware's convention before quoting angles. In his vectors, positive  $\theta$  moves the arm down.
- The velocity matrices  $A$  and  $B$  reuse the letters of the knee points  $A_i$ , the base points  $B_i$  and the constant  $b$ .
- Gap lead: Williams, Albus and Bostelman 2004, "3D cable-based Cartesian metrology system," *J. Robotic Systems* 21(5), 237–257 (the source of Appendix A).

## bentaleb2020improvement

**On the improvement of calibration accuracy of parallel robots – modeling and optimization.** T. Bentaleb and J. Iqbal, *Journal of Theoretical and Applied Mechanics*, vol. 58, no. 1, pp. 261–272, 2020. DOI 10.15632/jtam-pl/115863.

**Summary (3–4 lines):** The authors adapt two serial-robot kinematic calibration methods to a Delta robot. The forward method (FKM) minimises Cartesian position error; the inverse method (IKM) minimises each leg's constraint error. A 24-parameter model (8 per leg) is identified by nonlinear least squares using Jacobians of the models, in simulation only. Without noise both methods recover the parameters from 8 poses. With noise, IKM is much more robust, faster and converges from larger initial errors, but it needs 20–60 poses, with little gain beyond about 50.

### Key equations:

- Eq. (3.1), p. 264: FKM position error.  $X^r$  = real (measured) pose;  $X^{fkm}$  = pose from the forward model with parameters  $P$  and joint values  $\alpha$  (mm).

$$\partial X = X^r - X^{fkm}(P, \alpha)$$

- Eqs. (3.2)–(3.3), p. 264: IKM joint error for leg  $i$ .  $\alpha$  = actuated joint angle ( $^\circ$ ).

$$\alpha_i^{ikm} = F_{ikm}(P_i, X^r), \quad \Delta \alpha_i = \alpha_i^r - \alpha_i^{ikm}(P_i, X^r)$$

- Eq. (3.6), p. 265: the 8 parameters per leg ("Model 24").  $D_x$ ,  $D_y$  and  $D_z$  are base-joint offsets (mm),  $\theta$  and  $\alpha$  angles ( $^\circ$ ),  $La_x$  and  $La_y$  arm components (mm) and  $Lb$  the forearm length (mm).

$$P_i = [D_{x_i}, D_{y_i}, D_{z_i}, \theta_i, \alpha_i, La_{x_i}, La_{y_i}, Lb_i]$$

- p. 265 (text, unnumbered): identifiability condition.  $N_m$  = equations per pose (3 for FKM),  $N_c$  = poses,  $N_p$  = parameters. With  $N_p = 24$  this needs at least 8 poses.

$$N_m N_c \geq N_p$$

- Eq. (3.8), p. 265: IKM constraint for leg  $i$  at pose  $k$ .  $CB$  = vector from point  $C_i$  to point  $B_i$  (mm). As printed (see Unclear):

$$F_{ikm}^{i,k}(P_i, \chi_k^r, \alpha_{i,k}^r) = \left( \overline{CB}^{\backslash, r}_{i,k} \right)^{\mathrm{T}} \overline{CB}^{\backslash, r}_{i,k} - Lb_i$$

- Eqs. (3.11)–(3.12), p. 266: measured pose  $\chi$  and joint angle  $\alpha$  are the true values plus noise  $\varepsilon$ .

$$\chi_k^r = \chi_k^m + \varepsilon_{\chi, k}, \quad \alpha_k^r = \alpha_k^m + \varepsilon_{\alpha, k}$$

- Eq. (3.13), p. 266: the least-squares problems, over all parameters for FKM and per leg for IKM.

$$\min_{P=P^c} \sum_{k=1}^{N_c} \left( \varepsilon_k^{\mathrm{FKM}}(\chi_k^m, \alpha_k^m, P) \right)^2, \quad \min_{P_i=P_i^c} \sum_{k=1}^{N_c} \left( \varepsilon_{i,k}^{\mathrm{IKM}}(\chi_k^m, \alpha_{i,k}^m, P_i) \right)^2$$

- Eq. (3.14), p. 267: the implicit IKM constraint, differentiated.

$$\frac{\partial F_{ikm}^{k,i}}{\partial P_i} \Delta P_i + \frac{\partial F_{ikm}^{k,i}}{\partial \chi_k} \Delta \chi_k = 0$$

- Eqs. (3.16)–(3.18), p. 267: the FKM identification Jacobian.

$$[J_{\chi^k}]_{3 \times 3} [\Delta \chi^k]_{3 \times 1} = [J_{P^k}]_{3 \times 24} [\Delta P^k]_{24 \times 1}, \quad \Delta \chi^k = J_{k,P}^{\mathrm{fkm}} \Delta P, \quad J_{k,P}^{\mathrm{fkm}} = (J_{\chi^k})^{-1} J_{P^k}$$

$$J_{P^{fkm}} = \begin{bmatrix} -(J_{\chi^1})^{-1} J_{P^1} & \vdots & - \\ (J_{\chi^{N_c}})^{-1} J_{P^{N_c}} \end{bmatrix}$$

- Eq. (3.20), p. 268: one row of the IKM Jacobian (one leg, one pose). Eq. (3.21) stacks  $N_c$  rows into an  $N_c \times 8$  matrix.

$$J_{\{k, P_i\}^{ikm}} = \left[ \frac{\partial \epsilon_{i,k}^{ikm}}{\partial x_i}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial y_i}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial z_i}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial \theta_i}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial \alpha_i}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial L_{x_i}}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial L_{y_i}}, \frac{\partial \epsilon_{i,k}^{ikm}}{\partial L_b} \right]$$

- Eq. (4.1), p. 268: calibration gain  $G$  (%).  $P_r$  = real,  $P_n$  = nominal and  $P_c$  = calibrated parameters.

$$G = 100 \cdot \frac{||P_r - P_n|| - ||P_r - P_c||}{||P_r - P_n||}$$

### Figures / tables:

- Fig. 1, p. 263: delta schematic and parameter definitions ( $R_a$ ,  $R_b$ ,  $R$ ,  $L_a$ ,  $L_b$ ,  $\alpha_i$ ,  $\beta_i$ ,  $\theta_i$ ). Model dimensions:  $L_a = 260$  mm,  $L_b = 480$  mm, radii 194 and 30 mm.
- Fig. 2, p. 269: FKM with 60 poses and 0.01 mm initial error. (a) Per-parameter gain; several are negative, down to about  $-60$  %. (b) Pose errors before and after; they barely tighten.
- Fig. 3, p. 270: IKM with 60 poses and 1 mm initial error. (a) All gains positive, about 60–100 %. (b) Errors cluster tightly after calibration.
- Fig. 4, p. 271: IKM parameter error against number of poses (10–100), with 0.1 mm/ $^\circ$  parameter errors and 1  $\mu$ m noise. Errors drop sharply, then change little beyond about 50 poses.
- No tables. Results in the text, pp. 268–270:
  - No noise, 8 poses:  $G \approx 100$  % for both methods. From 1 cm initial error, FKM diverges and IKM converges ( $G = 99.9$  %).
  - FKM with 1  $\mu$ m and 0.01 $^\circ$  noise and 0.1 mm parameter error:  $G = -38.54$  % with 8 poses; 20 %, 38 % and 45.38 % with 20, 40 and 60.
  - IKM with 1 mm and 0.01 $^\circ$  errors:  $G = -6.17$  % with 8 poses; 62.54 %, 87.58 % and 86.5 % with 20, 40 and 60.

**Assumptions / limitations:**

- Simulation only; no physical robot or measurements.
- A purely geometric (kinematic) model: no joint compliance, backlash or load effects.
- Assumes an external device (e.g. a laser tracker) measures the pose.
- The noise levels ( $1\text{ }\mu\text{m}$ ,  $0.01^\circ$ ) are far below a hobby-servo robot's errors.

**Use in my report:**

- Section: II-D Calibration of low-cost and parallel robots. Claim: "Parallel-robot calibration usually identifies geometric parameters (24 for a Delta) from many externally measured poses by nonlinear least squares. The per-leg inverse formulation is more robust to noise than the forward one."
- Section: II-D / IV-J. Claim: "Kinematic calibration needs more poses as noise grows (20–60 here) and ignores non-geometric errors such as joint compliance."
- Section: II-E How this project differs. Claim: "Existing calibration methods rely on external metrology. This project corrects a non-geometric (compliance) error and checks the result in situ."
- Pointers from its literature review, supporting II-D: Bai 2015 (laser-tracker distance calibration of a Delta); Kamali 2016 (elasto-geometric calibration); Zhao 2015 (position error from 6 mm to under 1 mm).

**Unclear / check:**

- Eq. (3.8) subtracts  $L_{b\_i}$  from a squared length. The text says they use a difference of squares, so it should be  $L_{b\_i}^2$ . Don't copy it as printed.
- One block of Eq. (3.15) is written  $F\_mgi$ , which is leftover French notation (MGI) for  $F\_ikm$ .
- Signs disagree: Eq. (3.17) has no minus sign, but Eq. (3.18) has  $-(J\_X)^{-1}J\_P$ . Differentiating (3.14) gives the minus.
- The size printed as " $3 \times N\_c \times 24$ " in Eq. (3.18) means a  $3N\_c \times 24$  matrix.
- Parameter names clash on p. 263. The text gives " $R_b = 194\text{ mm}$  and  $R_n = 30\text{ mm}$ "; the nomenclature makes  $R_a$  the base radius and  $R_b$  the platform's; Fig. 1 labels  $OA\_i = R_b$  and  $C\_iP = R_a$ . So 194 mm is presumably the base radius.
- The abstract says they studied the workspace boundary and how parameter errors change it. The body has no such results.
- The Discussion (p. 270) gives a forward-method gain of 48.27 %, which matches nothing in Section 4.

- p. 270 says "measurement errors in the order of 1 mm and 0.01 degree on the parameters". It's unclear whether 1 mm is noise or the initial parameter error.
- In the review, Nubiola and Bonev's mean error "0.873 to 0.479" has no units (probably mm).
- Gap leads: Bai et al. 2015, *Proc. IMechE Part C* 230(3), 414–424; Kamali et al. 2016, ICRA, 4320–4327; Zhao et al. 2015, *Int. J. Adv. Robotic Systems* 12(9), 1–13; Nguyen et al. 2015, *Neurocomputing* 151, 996–1005 (EKF + ANN for non-geometric error). Check the last against the "Nguyen 2022" in your outline (IV-J).

tirathompson2008digital

**Digital Servo Calibration and Modeling.** E. Tira-Thompson, writing qualifier (unpublished report), 15 Dec. 2008, 8 pp. No institution, venue or URL is printed. It uses the Tekkotsu framework, so it is probably from Carnegie Mellon University, but the report doesn't say. URL and access date still needed.

**Summary (3–4 lines):** The author tests Robotis Dynamixel AX-12 digital servos (Bioloid kit, 12 V supply) with cheap methods, mainly a camera tracking a strip of tape on the horn. Position feedback matched the camera closely: aggregate slope 1.0014, and an unloaded  $\sigma$  of  $0.36^\circ$  from the target. So no global position calibration was needed. Speed command and feedback needed linear fits; the command fits vary between servos, the feedback fits don't. Under load the servo deflects roughly in proportion to torque, with separate models for lifting and lowering because gear friction changes sign. Feeding the predicted deflection forward as an offset made the motion much more accurate (Fig. 16).

**Key equations** (none are numbered; PDF page given):

- Table 2, p. 3: position fit.  $P_{vis}$  = camera-measured angle,  $P_{Fdbk}$  = servo feedback angle ( $^\circ$ ).

$$P_{vis} = x \cdot P_{Fdbk}$$

- Table 3, p. 5: speed-command fit.  $S_{cmd}$  = command and  $S_{tgt}$  = target speed ( $^\circ/s$ );  $x$  and  $b$  are fitted.

$$S_{cmd} = x \cdot S_{tgt} + \operatorname{sign}(S_{tgt}) \cdot b$$

- Table 4, p. 5: speed-feedback fit.  $S_{actual}$  = measured speed,  $S_{Fdbk}$  = reported speed.

$$S_{\text{actual}} = x \cdot S_{\text{Fdbk}}$$

- p. 6: deflection (position error) while moving, against required torque and speed. Units are implied by the figures, not stated: defl in °, torque in kgf·cm, speed in °/s.

```
\begin{aligned}
\text{defl}_{\text{up}} \&= -0.138 \cdot \text{torque} + 0.0713 \cdot \text{speed} \\
\text{defl}_{\text{down}} \&= -0.0520 \cdot \text{torque} + 0.0777 \cdot \text{speed}
\end{aligned}
```

- p. 7: the minimum command offset needed to break static friction. acc = acceleration; mass in kg; radius = lever arm.

```
\begin{aligned}
\text{defl}_{\text{up}} \&= 0.412 \cdot \text{torque} + 0.137 \cdot \text{speed} - 0.0114 \cdot \text{acc} - \\
&5.36 \cdot \text{mass} \cdot \text{radius} \\
\text{defl}_{\text{down}} \&= -0.514 \cdot \text{torque} + 0.182 \cdot \text{speed} - 0.0229 \cdot \text{acc} - \\
&3.23 \cdot \text{mass} \cdot \text{radius}
\end{aligned}
```

- p. 7: inverting the deflection model to estimate a carried mass.  $r$  = lever radius,  $g$  = gravity,  $\theta$  = arm angle,  $\ddot{\theta}$  = angular acceleration;  $a$  and  $b$  are the fitted coefficients,  $a'$  and  $b'$  the matching constants; load = the servo's load feedback.

```
\text{defl} = a \cdot \text{torque} + b \cdot \text{speed}, \text{torque} = \frac{1}{a} \cdot \text{defl} - \frac{b}{a} \cdot \text{speed}
```

```
\text{mass} = a' \cdot \frac{\text{defl}}{r \cdot g \cdot \cos \theta + r^2 \cdot \ddot{\theta}} - b' \cdot \frac{\text{speed}}{r \cdot g \cdot \cos \theta + r^2 \cdot \ddot{\theta}}
```

```
\text{mass} = \frac{-5.24 \cdot \text{defl} + 0.346 \cdot \text{speed} - 0.245 \cdot \text{load}}{r \cdot g \cdot \cos \theta + r^2 \cdot \ddot{\theta}}
```

### Figures / tables:

- Table 1, p. 2: AX-12 published specs. Holding torque 12.0 kgf·cm at 7 V and 16.5 kgf·cm at 10 V; no-load speed 0.269 and 0.196 s/60°.
- Unnumbered figure, p. 2: the controller's compliance curve, showing margin, slope, punch and torque limit against position error.
- Figs. 1–5, p. 3: the camera rig. The rotation centre is found where line bisectors intersect ( $\sigma = 0.21$  px).

- Table 2, p. 3: position-fit slope per servo, 0.9980–1.0101. Fit  $\sigma$  0.42–1.63°;  $\sigma$  from target 0.30–0.36°.
- Fig. 6, p. 4: position fit and its residuals (within  $\pm 4^\circ$ ).
- Figs. 7–9, p. 4: the dead zone in free spin, and uncalibrated vs calibrated speed. Slow rotations stall because of gear friction.
- Fig. 10, Table 3 and Table 4, p. 5: command fits  $x = 0.671\text{--}0.776$  and  $b = 7.5\text{--}15.7$  °/s. Feedback slopes 2.05–2.08 ( $\sigma$  of fits 0.0133).
- Figs. 12–16, p. 6: torque rig with masses 0–500 g, radii 75–150 mm and periods 2–8 s.
  - Fig. 14: load and deflection (about 2–10°) both rise with required torque.
  - Fig. 15: predicted vs measured deflection.
  - Figs. 13 and 16: commanded vs measured position, before and after the deflection prediction is applied.
- Fig. 17, p. 7: free-spin speed used as torque control (about 34.9 °/s per kgf·cm). Fig. 18: predicted static-friction offset.
- Figs. 19–20, p. 8: mass estimated from servo feedback.

#### **Assumptions / limitations:**

- The Dynamixel AX-12 is a digital servo with position, speed and load feedback over a serial bus. It is not an open-loop PWM hobby servo like yours, so its accuracy figures don't transfer.
- Tested at one supply voltage (12 V). Voltage and temperature effects weren't studied (Future work, p. 8).
- Not peer-reviewed, and only seven servos.
- The models' units aren't stated.

#### **Use in my report:**

- Section: II-D Calibration of low-cost and parallel robots. Claim: "Low-cost servos can be calibrated without precision rigs, e.g. with a camera. Under load they deflect from the target roughly in proportion to torque, and adding the modelled deflection as an offset improves accuracy."
- Section: III-F Firmware. Claim: "Gear friction makes a servo's position error depend on the direction of motion (separate lifting and lowering models)." This supports the upward final approach.
- Section: III-H Calibration and position correction, supporting. Claim: "A linear torque-to-deflection model, i.e. a joint stiffness, can be used as a feed-forward correction."

- Section: IV-B Joint calibration, comparison only. A feedback digital servo was within  $0.36^\circ$  of target unloaded; your open-loop servos went from  $3.4^\circ$  to  $0.5^\circ$  rms after recalibration.

#### Unclear / check:

- The Table 2 caption says the values are in milliradians, but the column headers and the text ( $0.36^\circ$ ) use degrees. Use degrees.
- The text (p. 8) says the AX-12 is rated 12 kgf·cm at 12 V. Table 1 gives 12.0 kgf·cm at 7 V and 16.5 kgf·cm at 10 V.
- The p. 6 deflection models have negative torque coefficients, but Fig. 14 plots positive deflections. The sign convention isn't stated, so check it before quoting a direction.
- The static-friction models on p. 7 reuse the names `defl_up` and `defl_down` for a different quantity.
- The report quotes the Hitec manual (communication errors about 10 % of the time). If you use that point, cite the Hitec document.
- Gap lead: ref. [4] Lange and Hirzinger 1996, "Learning force control with position controlled robots," ICRA.

#### iso1998manipulating

**ISO 9283:1998 — Manipulating industrial robots — Performance criteria and related test methods.** International Organization for Standardization, 2nd ed., Apr. 1998, 60 pp.

Committee ISO/TC 299; ICS 25.040.30. Your file is the ISO catalogue page,

<https://www.iso.org/standard/22244.html> (accessed 9 Oct. 2026), not the standard itself.

**Summary (3–4 lines):** The page gives the standard's bibliographic data and status. It was last reviewed and confirmed in 2021, so this edition is still current, though the page also shows stage 90.60 (under review). It replaced ISO 9283:1990 and its 1991 amendment, and costs CHF 204. The page has no technical content: no definitions of pose accuracy or repeatability, no test poses and no formulas.

**Key equations:** none (not on the page).

**Figures / tables:** none.

#### Assumptions / limitations:

- A catalogue page only. It supports the citation details and the standard's current status, and nothing technical.

- The status is shown two ways: "Close of review [90.60]" and "Stage 90.60 (Under review)".

#### Use in my report:

- Section: II-F Performance metrics and benchmarks / IV-A Performance metrics. Claim: "Position accuracy and repeatability follow the definitions of ISO 9283:1998 (confirmed in 2021)." This page backs the citation only; the definitions must come from the standard itself.

#### Unclear / check:

- Get the full text (VUW Library or Standards NZ) to state the accuracy and repeatability definitions and their clause numbers. Don't write the formulas from memory.
- The standard is under review. Recheck the page before submission, since a new edition would change the citation.
- IEEE format: *Manipulating Industrial Robots — Performance Criteria and Related Test Methods*, ISO 9283:1998, 1998.
- The outline also mentions ISO 230-2; nothing in this batch covers it.

#### pollak2023simulation

**Simulation Design and Measurement of Welding Robot Repeatability Utilizing the Contact Measurement Method.** M. Pollák and K. Goryl, *Machines*, vol. 11, no. 7, Art. no. 734, Jul. 2023 (15 pp.; published 13 Jul. 2023). DOI 10.3390/machines11070734. Cite key provisional.

**Summary (3–4 lines):** The authors checked the unidirectional position repeatability of a Panasonic TM-2000 six-axis welding robot in a factory cell. They modelled the cell in RoboDK and placed a 1200 mm ISO cube where the robot welds most. Five points (P1–P5) on the cube's bottom face were each measured for 30 cycles with a digital dial indicator (INSIZE 2112-254F, 0.001 mm step) touched by the torch nozzle; X and Y were measured separately and Z was dropped. Using  $\pm 3\sigma$  per axis, every point beat the maker's  $\pm 0.10$  mm (worst was P1: 0.086 mm in X, 0.082 mm in Y). The worst point was the centre of the plane, not the outer points as they expected.

#### Key equations:

- Eq. (1), p. 5: ISO 9283 unidirectional pose accuracy.  $\bar{x}$ ,  $\bar{y}$ ,  $\bar{z}$  = barycentre of the attained positions over  $n$  cycles (mm);  $x_c$ ,  $y_c$ ,  $z_c$  = commanded position (mm);  $AP_x$ ,  $AP_y$ ,  $AP_z$  = per-axis offsets (mm).

$$AP\_P = \sqrt{(\bar{x} - x_c)^2 + (\bar{y} - y_c)^2 + (\bar{z} - z_c)^2} = \sqrt{(AP\_x)^2 + (AP\_y)^2 + (AP\_z)^2}$$

- Eq. (2), p. 5: barycentre.  $x_j, y_j, z_j$  = j-th attained position (mm);  $n$  = number of cycles.

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j, \quad \bar{y} = \frac{1}{n} \sum_{j=1}^n y_j, \quad \bar{z} = \frac{1}{n} \sum_{j=1}^n z_j$$

- Eq. (3), p. 5: ISO 9283 unidirectional position repeatability.  $\Gamma$  = mean distance of the attained positions from the barycentre (mm);  $S_l$  = sample SD of those distances (mm).

$$RP_l = \bar{l} + 3S_l$$

- Eq. (4), p. 5: definitions for Eq. (3). The "+" signs inside  $l_j$  are missing in the paper and are restored here (see Unclear).

$$\bar{l} = \frac{1}{n} \sum_{j=1}^n l_j, \quad l_j = \sqrt{(x_j - \bar{x})^2 + (y_j - \bar{y})^2 + (z_j - \bar{z})^2}, \quad S_l = \sqrt{\frac{\sum_{j=1}^n (l_j - \bar{l})^2}{n-1}}$$

- Eq. (5), p. 12: the authors' per-axis substitute, used because the jig was moved between points and the axes were measured separately.  $P_{nj}$  = j-th reading at point  $n$  along one axis (mm);  $\bar{P}_n$  = mean of those readings (mm).

$$RP_{\{P_n\}} = \pm 3 S_{\{P_n\}} = \pm 3 \sqrt{\frac{\sum_{j=1}^n (P_{nj} - \bar{P}_n)^2}{n-1}}$$

### Figures / tables:

- Fig. 1, p. 1: four targets showing low/high accuracy against low/high repeatability.
- Fig. 2, p. 4: an example ISO cube with diagonal plane C1-C2-C7-C8 and points P1-P5.
- Fig. 3, p. 4: accuracy AP (commanded point O to barycentre G) and repeatability  $RP_l$  (sphere about G).
- Table 1, p. 4: ISO pose-test conditions. 100 % load and 100 % speed are required; 10 % load and 50 %/10 % speed are optional; 5 points (P1-P5) and 30 cycles.
- Fig. 4, p. 6: TM-2000 specs: 6 kg payload, 1999 mm reach, 6 axes, 180 m/min maximum speed,  $\pm 0.10$  mm repeatability, 216 kg.
- Fig. 5, p. 6: the dial indicator on its jig, touched by the welding nozzle.
- Fig. 6, p. 7: the cube used; measurement plane C5-C6-C7-C8 (bottom face), P1 at the centre.

- Table 2, p. 7: test conditions used: 90 m/min (50 % of maximum), 3 kg (50 % of maximum), P1–P5, 30 cycles, X and Y axes.
- Figs. 7–9, pp. 8–9: the real cell, the RoboDK model and the cube placed in it.
- Fig. 10, p. 10: X- and Y-axis measurement at P1.
- Table 3, pp. 10–11: 30 X-axis readings per point (mm).
- Table 4, p. 11: 30 Y-axis readings per point (mm).
- Table 5, p. 12:  $\pm 3S$  results. X: 0.086, 0.030, 0.037, 0.027, 0.062 mm. Y: 0.082, 0.029, 0.053, 0.025, 0.033 mm (P1–P5). All are below 0.1 mm. I recomputed all ten values from Tables 3–4 and they match.
- Figs. 11–13, pp. 12–13: bar charts of Table 5 against the declared 0.1 mm.

### Assumptions / limitations:

- This isn't a full ISO 9283 RP test. Each axis was measured alone, Z was dropped, and Eq. (5) leaves out the mean-distance term  $\bar{r}$ , so the figures understate  $RP_I$ .
- Load and speed were 50 % and 50 %. That's neither the required condition (100 %/100 %) nor the optional one (10 % load) in their own Table 1.
- It's a contact method. The indicator spring and a 10 mm pre-compression push on the nozzle, and every touch comes from one direction.
- One robot and one session. The warm-up was an 8 h shift, and ambient was 21 °C and 38 % RH.

### Use in my report:

- Section: II-F Performance metrics and benchmarks. Claim: "ISO 9283 defines pose accuracy AP as the distance from the commanded position to the barycentre of the attained positions, and repeatability as  $RP_I = \bar{r} + 3S_I$ , the radius of a sphere about that barycentre." This is a secondary source; cite the standard itself as the primary once you have the full text.
- Section: IV-A Performance metrics, supporting. Their ISO test layout (5 points on a cube plane, 30 cycles) is a reference for describing how the project's test was scaled down.
- Section: IV-D Position accuracy, repeatability and drift. Comparison claim: "An industrial welding arm specified at  $\pm 0.1$  mm measured 0.03–0.09 mm per axis; this project's  $\sim 2$  mm repeatability is about 20–70 times larger, as expected for open-loop hobby servos." Say that the metrics differ: their per-axis  $\pm 3S$  vs your  $RP_I$ .
- Section: IV-J Critical analysis, supporting. Simplified per-axis  $\pm 3\sigma$  figures aren't comparable with ISO  $RP_I$ , so the report should say which one it uses.

**Unclear / check:**

- Eq. (4), p. 5: the "+" signs between the squared terms in  $I_j$  are missing. The sentence after Eq. (3) is also garbled (it calls  $S_I$  "the standard deviation  $\bar{x}$ ,  $\bar{y}$ ,  $\bar{z}$ ").
- Table 1, p. 4: the speed entry "50%—10% optional" is ambiguous. Check it against the standard.
- Table 3 shows a trend that  $\pm 3S$  hides (my reading). At P4 in X the readings drift from  $-0.018$  to  $-0.056$  mm over 30 cycles. At P1 in X the mean is  $+0.096$  mm after zeroing on the first touch.
- p. 2 says that more repetitions give worse repeatability, with no support.
- p. 9 says ISO 9283 requires  $20\text{--}22$  °C ambient. Check the standard's wording before repeating this.
- Wording: "digital aberration meter" (p. 9) and "deflection meter" both mean the dial indicator.
- Gap leads: Slamani et al. 2012 (ref. [13]), ISO 9283 tests with a laser tracker, interferometer and ballbar. Vocetka et al. 2021 (ref. [15]), thermal drift and repeatability compensation (for IV-D drift). Józwik et al. 2016 (ref. [14]), camera-based repeatability, where the measured RP was worse than the declared 0.06 mm.

**uscategui2023highprecision**

**High-Precision Magnetic Testbed Design and Simulation for LEO Small-Satellite Control Test.** J. Uscategui, X. Wang, G. Cuba and M. Guarirapa, *Aerospace*, vol. 10, no. 7, Art. no. 640, Jul. 2023 (24 pp.; published 15 Jul. 2023). DOI 10.3390/aerospace10070640. Cite key provisional.

**Summary (3–4 lines):** Beihang University designed, simulated (Ansys Maxwell) and built IS501NMTB, a three-axis square Helmholtz-coil testbed. The coils have  $\sim 1.03$  m sides with 0.5 m between each pair, on a wooden frame. It replays the geomagnetic field along a LEO orbit (Lilacsat-1 elements, SGP4 propagation, WMM2020). A fluxgate magnetometer at the centre closes a PID loop on three programmable DC supplies, and the magnetometer was calibrated for hard iron, soft iron, scale, non-orthogonality and offset. Over 1080 orbit samples, the field at the centre tracked the target within about 0.11 % per axis (they report this as " $>99.89$  % accuracy"). An axial scan before calibration gave  $\leq 0.12$  % deviation over  $\pm 13$  cm, and simulation gave a uniform region of about  $28 \times 25 \times 25$  cm.

**Key equations:**

- Eq. (1), p. 3: J2 drift of the RAAN.  $\Omega, \Omega_0$  = current and initial RAAN (rad);  $n$  = mean motion (rad/s);  $R_e$  = Earth's mean radius (m);  $P$  = semilatus rectum (m);  $J_2$  = second zonal harmonic (dimensionless);  $i$  = inclination;  $\Delta t$  = elapsed time (s).

$$\dot{\Omega} = \dot{\Omega}_0 - 1.5 \frac{n}{P^2} J_2 \cos(i) \Delta t$$

- Eq. (2), p. 3: J2 drift of the argument of perigee  $\omega$  (rad);  $\omega_0$  = initial value.

$$\dot{\omega} = \dot{\omega}_0 + 0.75 \frac{n}{P^2} J_2 \left( 5 \cos^2(i) - 1 \right) \Delta t$$

- Eq. (3), p. 3: mean anomaly  $M$  with the J2 correction;  $M_0$  = initial value (rad);  $e$  = eccentricity. The printed brackets don't balance. This is the standard form; see Unclear.

$$M = M_0 + \left[ n + 0.75 \frac{n}{P^2} J_2 \left( 2 - 3 \sin^2(i) \right) \sqrt{1-e^2} \right] \Delta t$$

- Eq. (4), p. 3: orbit radius.  $R_{\text{sat}}$  = distance from Earth's centre (m);  $a$  = semi-major axis (m);  $\theta$  = true anomaly.

$$R_{\text{sat}}(\theta) = \frac{a(1-e^2)}{1 + e \cos(\theta)}$$

- Eq. (5), p. 3: true anomaly  $\theta$  from eccentric anomaly  $E$ .

$$\tan\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+e}{1-e}} \tan\left(\frac{E}{2}\right)$$

- Eq. (6), p. 3: Kepler's equation.

$$M = E - e \sin(E)$$

- Eq. (7), p. 4: WMM main field as minus the gradient of a scalar potential.  $B_m$  = main field (nT);  $\lambda$  = longitude;  $\varphi'$  = geocentric latitude;  $r$  = radius (m);  $t$  = time.

$$B_m(\lambda, \varphi', r, t) = -\nabla V(\lambda, \varphi', r, t)$$

- Eq. (8), p. 4: spherical-harmonic expansion, as printed. "sen" is sin. The left side should be  $V$ , not  $\nabla V$  (see Unclear).  $N = 12$  for WMM;  $a = 6371200$  m geomagnetic reference radius;  $g, h$  = Gauss coefficients (nT);  $P_n^m$  = Schmidt semi-normalised associated Legendre functions.

$$\nabla V(\lambda, \varphi, r, t) = a \sum_{n=1}^N \left( \frac{a}{r} \right)^{n+1} \sum_{m=0}^n \left[ g_n^m(t) \cos(m\lambda) + h_n^m(t) \sin(m\lambda) \right] P_n^m(\varphi)$$

- Eq. (9), p. 4: radial component.  $\theta$  = colatitude;  $\phi$  = longitude;  $R$  = Earth's radius;  $k$  = maximum degree.

$$B_r = -\frac{\partial V}{\partial r} = \sum_{n=1}^k \left( \frac{R}{r} \right)^{n+2} (n+1) \sum_{m=0}^n \left[ g_n^m \cos(m\varphi) + h_n^m \sin(m\varphi) \right] P_n^m(\theta)$$

- Eq. (10), p. 4: colatitude component.

$$B_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = -\sum_{n=1}^k \left( \frac{R}{r} \right)^{n+2} \sum_{m=0}^n \left[ g_n^m \cos(m\varphi) + h_n^m \sin(m\varphi) \right] \frac{\partial P_n^m(\theta)}{\partial \theta}$$

- Eq. (11), p. 4: longitude component.

$$B_\phi = -\frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} = -\frac{1}{\sin \theta} \sum_{n=1}^k \left( \frac{R}{r} \right)^{n+2} \sum_{m=0}^n m \left[ -g_n^m \sin(m\varphi) + h_n^m \cos(m\varphi) \right] P_n^m(\theta)$$

- Eq. (12), p. 5: Biot–Savart law.  $B$  = flux density (T);  $I$  = current (A);  $d\mathbf{l}$  = current element (m);  $r$  = distance to the field point (m);  $\hat{r}$  = unit vector;  $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ .

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \hat{r}}{r^2}$$

- Eq. (13), p. 5: on-axis field of a pair of **square** coils of side  $L = 2a$ , centred at  $z = \pm d/2$ .  $B_z$  (T);  $a$  = half-side (m);  $N$  = turns;  $I$  (A);  $d$  = coil separation (m);  $z$  = axial position from the centre (m). Exponent placement as I read it: the  $\frac{1}{2}$  power applies to the  $(2a^2 + \dots)$  factor only, which is the correct square-loop form. Verify on the PDF.

$$B_z = \frac{2\mu_0 I a^2 N}{\pi} \left[ \frac{1}{\left( a^2 + \left( z + \frac{d}{2} \right)^2 \right)^{3/2}} - \frac{1}{\left( a^2 + \left( z - \frac{d}{2} \right)^2 \right)^{3/2}} \right]$$

- Eq. (14), p. 6: on-axis field of a **circular** Helmholtz pair of radius  $a$ , to fourth order ( $N = 1$ ).

$$B_z(\rho=0; z) = \frac{8\mu_0 I}{5\sqrt{5}a} \left[ 1 - \frac{144}{125} \left( \frac{z}{a} \right)^4 \right]$$

- Eq. (15), p. 6: radial field near the axis ( $\rho \ll a$ ) of a circular pair;  $\rho$  = radial distance (m).

$$B_\rho(\rho, z)_{\rho \ll a} = \left( \frac{8\mu_0 I}{5\sqrt{5}a} \right) \left( \frac{288}{125} \right) \left( \frac{\rho}{a} \right) \left( \frac{z}{a} \right)^3$$

- Eqs. (16)–(18), p. 6: absolute, relative and percentage error.  $R^*$  = measured value;  $R$  = reference value.

$$\Delta R = R^* - R, \quad e = \frac{|R^* - R|}{R}, \quad e_{\%} = \frac{|R^* - R|}{R} \times 100\%$$

- Eqs. (19)–(23), p. 7: magnetometer error terms.  $b_{hi}$  = hard-iron offset vector (nT; the text calls it soft iron, see Unclear);  $A_{si}$  = soft-iron matrix (dimensionless);  $S$  = scale-factor matrix;  $M$  = non-orthogonality matrix, with  $\varepsilon_x, \varepsilon_y, \varepsilon_z$  the sensor-axis directions in the coil frame;  $b_{so}$  = sensor offset (nT).

$$\begin{aligned} b_{hi} &= [b_{hix} \ b_{hiy} \ b_{hiz}]^T, \quad \text{quad} \\ A_{si} &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{32} & a_{33} \end{bmatrix}, \quad \text{quad} \\ S &= \text{diag}(S_x \ S_y \ S_z), \quad \text{quad} \\ M &= [\varepsilon_x \ \varepsilon_y \ \varepsilon_z]^{-1}, \quad \text{quad} \\ b_{so} &= [b_{sox} \ b_{soy} \ b_{soz}]^T \end{aligned}$$

- Eqs. (24)–(26), p. 7: calibration model.  $h$  = calibrated field (nT);  $h_m$  = measured field (nT);  $A$  = combined matrix;  $b$  = combined offset (nT).

$$h = A^{-1} (h_m - b), \quad \text{quad} \quad A = S M A_{si}, \quad \text{quad} \quad b = S M b_{hi} + b_{so}$$

- Calibration result, p. 15 (unnumbered): fitted  $A$  and  $b$  from 500 points at the cage centre, referenced to the 54 813.1 nT local field.  $A^{-1}$  is also printed;  $b$  is in nT.

$$A = \begin{bmatrix} 1.2243 & -0.0127 & -0.0086 \\ -0.0127 & 1.2330 & 0.0109 \\ -0.0086 & 0.0109 & 1.1539 \end{bmatrix}, \quad \text{quad} \quad b = [-334.775002, -11.989813, 572.421773]^T$$

- Eq. (27), p. 13: zero-calibration command.  $B_d$  = desired field;  $B_s$  = field sensed in the cage;  $B_z$  = coil field from Eq. (13) (all nT).

$$B_d = B_s \times (-1) + B_z$$

- Eqs. (28)–(30), p. 14: coil current for a desired field. These are **circular-coil** terms (see Unclear). The paper's  $L$  here is not the side length  $L$ .

$$I = \frac{2B_z}{\mu_0 a^2 N \sqrt{K + L}}, \quad K = \frac{1}{\left(a^2 + \left(z + \frac{d}{2}\right)^2\right)^{3/2}}, \quad L = \frac{1}{\left(a^2 + \left(z - \frac{d}{2}\right)^2\right)^{3/2}}$$

### Figures / tables:

- Table 1, p. 3: Lilacsat-1 orbit (500 km altitude, 97.41° inclination,  $e = 0.0012$ ) and the testbed location (39.98° N, 116.35° E).
- Fig. 1, p. 5: Helmholtz pair geometry (side  $L$ , half-side  $a$ , separation  $d$ , axis  $z$ ).
- Fig. 2, p. 6: simulated "stable field" sphere at the cage centre around a small satellite.
- Fig. 3, p. 8: ground track coloured by field magnitude (nT), 17 May 2021 04:00–07:00 UTC.
- Fig. 4, p. 8: field components along the orbit, about  $\pm 50\,000$  nT.
- Table 2, p. 9: coil pairs. Sides 1.03/1.02/1.03 m; half-sides 0.515/0.51/0.515 m; separation 0.5 m on every axis; turns 97.18/111.2/59.12 ("including connection lines").
- Fig. 5, p. 9: control architecture. Desired field  $\rightarrow$  PID  $\rightarrow$  DC supply  $\rightarrow$  coils  $\rightarrow$  magnetometer feedback, per axis.
- Table 3, p. 10: zero-calibration settings. X  $-8.115$  V,  $-0.5715$  A,  $14.2\,\Omega$ . Y  $-0.278$  V,  $-0.041$  A,  $6.87\,\Omega$ . Z  $-9.98$  V,  $-1.082$  A,  $9.23\,\Omega$ .
- Fig. 6, p. 10: 3D model of the three coil pairs and the structure.
- Fig. 7, p. 11: simulated  $|B|$  for the Y-axis coils in a horizontal plane (scale 0–23  $\mu$ T).
- Figs. 8–10, pp. 11–12: simulated Y-coil field along Y, X and Z. It's flat at about 2.38  $\mu$ T. Marker pairs differ by 0.0034  $\mu$ T over 0.258 m (Y), 0.0003  $\mu$ T over 0.283 m (X) and 0.0002  $\mu$ T over 0.256 m (Z).
- Fig. 11, p. 12: the built testbed: wooden frame, aluminium supports and plastic connectors.
- Fig. 12, p. 13: each coil is split into sub-coils joined by jumper ports so the turns can be changed.

- Fig. 13, p. 13: DC supplies. ITECH IT6433 ( $\pm 60$  V,  $\pm 2.5$  A) on X and Z; IT6431 ( $\pm 15$  V,  $\pm 10$  A) on Y.
- Fig. 14, p. 14: control software showing the sensed field (31 020, 2067, 36 220 nT) and currents.
- Fig. 15, p. 15: Simulink module for orbit, field and testbed control.
- Fig. 16, p. 16: magnetometer point cloud before and after calibration.
- Table 4, p. 16: seven orbit samples comparing predicted and reproduced X, Y, Z field (nT).
- Table 5, p. 17: the same seven samples with per-axis "accuracy" (99.80–99.99 %) and error (–0.18 to +0.20 %).
- Table 6, p. 17: averages over 1080 samples. Accuracy 99.893/99.895/99.889 %; error 0.107/0.105/0.111 %; SD 0.0846/0.0799/0.0903 %.
- Figs. 17–20, p. 18: reproduced field against target along the orbit (all axes, then X, Y, Z).
- Fig. 21, p. 19: per-axis error over time, inside about  $\pm 0.12$  %.
- Table 7, p. 19: measured field along the X and Y axes at  $\pm 13$  cm, before calibration. X: 9863–9875 nT at  $-0.080$  A; Y:  $-4307$  to  $-4316$  nT at  $-0.0573$  A. Deviation from the centre is  $\leq 0.12$  %.
- Table 8, p. 20: comparison with five other testbeds (size, error 0.2–4 %, currents, software). "This work" has  $L = 1$  m, 0.11 % error and 1.7 A.

### Assumptions / limitations:

- Uniformity comes from simulation plus one axial line per axis (Table 7, 9 points). There's no 3D map of the working volume.
- The ">99.89 %" figure compares the commanded field with the field read by the same centre magnetometer that closes the control loop. So it measures tracking at one point, not independent accuracy or uniformity.
- They assume the stable region is a sphere of diameter  $d/2$  at the centre (p. 6, citing [18], [19]).
- The coils were treated as ideal thin loops in the formulas, though they're built from spread-out sub-coils.

### Use in my report:

- Section: I-A Problem statement. Claim: "Helmholtz-cage testbeds for small satellites need a known uniform region; a 1 m square-coil cage gives roughly 0.1 % uniformity over a  $\sim 25$  cm diameter, and builders check it by simulation and a few axial points rather than a full 3D map." This motivates automated 3D mapping.

- Section: II-A Magnetic field mapping systems. Claim: "Cage builders typically verify uniformity with simulation plus manual measurements along the coil axes (9 points per axis in IS501NMTB, Table 7), not a volumetric map." Pairs with Wilson's hand map as the baseline.
- Section: IV-K Limitations / VI Future work. Claim: "The on-axis field of a square Helmholtz pair is given by Eq. (13), for the planned comparison with Wilson's square coils." Use with the caveats under Unclear.
- Section: IV-K Limitations. Claim: "A magnetometer needs hard-iron (offset), soft-iron (matrix), scale-factor and non-orthogonality correction,  $h = A^{-1}(h_m - b)$  (Eqs. (24)–(26))." Take the definitions from a primary source (Renaudin 2010).
- Section: II-B Positioning in magnetic environments, supporting. Claim: "Magnetic testbeds are framed in wood with aluminium and plastic fittings to stay non-magnetic (p. 12)." This matches the project's wooden frame.
- Section: IV-G, context only. Their local field was 54 813.1 nT in Beijing (p. 15), which isn't an NZ value.

#### Unclear / check:

- p. 5 says uniform fields need the separation  $d$  equal to the side length  $L$ . That's wrong for square coils. My calculation with Eq. (13) puts the optimum at  $d \approx 0.5445 L$ ; get a primary source before you quote any spacing. Their own build (0.5 m for 1.03 m sides, about 0.49  $L$ ) contradicts their text.
- Formulas are mixed. Eq. (13) is for square coils, but Eqs. (14)–(15) and (28)–(30) are circular-coil formulas, and the currents come from Eq. (28).
- Table 7 is better than ideal theory allows (my calculation, ideal thin loops). For  $a = 0.515$  m and  $d = 0.5$  m, Eq. (13) gives about 1.25 % on-axis drop at 12.5 cm; even at the optimum spacing it's about 0.27 %. The sub-coil winding (Fig. 12) may explain part of the gap. Treat the  $\leq 0.12$  % figure with caution.
- The text on p. 19 refers to these currents as giving "the results in Table 6" but means Table 7. Eq. (18) defines  $e\%$  as an absolute value, yet Table 7 lists signed values.
- The simulated field (2.38  $\mu\text{T}$ , Figs. 8–10) is about 20 times smaller than Earth's field, and the simulation current isn't stated.
- On p. 7 the hard- and soft-iron descriptions are swapped. The text calls the offset vector  $b_{hi}$  "soft iron" and the matrix  $A_{si}$  a hard-iron "bias"; the subscripts are right. Eq. (20) has  $a_{32}$  where the symmetric form needs  $a_{23}$ .
- p. 6 calls  $e\%$  "permeability"; it's a percentage error.
- Eq. (8): the left side should be  $V$ , not  $\nabla V$ . Eq. (3): the brackets don't balance as printed.

- Table 2 gives non-integer turns (97.18, 111.2, 59.12) with no explanation.
- Table 3 and Fig. 14 disagree: Y current  $-0.041$  A vs  $-0.020$  A, and Z  $-1.082$  A vs  $-1.075$  A. For Y,  $V/I = 6.78 \Omega$ , but  $6.87 \Omega$  is listed.
- Table 5, row 3: "1,5293.10" is a typo for 15 293.10.
- Gap leads: Batista et al. 2018, *IEEE TAES* 54, 392–403 (ref. [19]), three-axis Helmholtz coil design and validation, possibly square coils. Not yet added to Flags because it may duplicate the existing "square-coil Helmholtz paper" task. da Silva et al. 2019 (ref. [32]). Piergentili et al. 2011 (ref. [33]). Renaudin et al. 2010 (ref. [29]), magnetometer calibration. Wang et al. 2019 and 2018 (refs. [26], [27]), dipole localisation and position error from field and gradient. Yue et al. 2021 (ref. [28]), calibrating the non-orthogonality and scale-factor error of a triaxial Helmholtz coil.

## cervettini2021development

### Development and Experimentation of a CubeSat Magnetic Attitude Control System

**Testbed.** G. Cervettini, S. Pastorelli, H. Park, D. Y. Lee and M. Romano, *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 2, pp. 1345–1350, Apr. 2021 (Correspondence; published online 24 Nov. 2020). DOI 10.1109/TAES.2020.3040032. Cite key provisional.

**Summary (3–4 lines):** The authors built three air-core magnetorquers (3D-printed frames, AWG34 wire) and fitted them to the Naval Postgraduate School's CubeTAS air-bearing hemisphere, which floats inside a three-axis Helmholtz cage. The magnetorquers' torque is too small to beat the testbed's own disturbances: mass imbalance, friction and air drag. So they drove the cage to augment the field (up to  $\pm 2$  G per axis). With 1.705 G applied, a B-dot controller stopped a yaw-only spin in 81 s (simulation: 77 s) and three-axis tumbling in 156 s. The cage serves only as a field source; its uniformity isn't measured or reported.

### Key equations:

- Eq. (1), p. 1347: quaternion kinematics,  $\dot{q} = R(q)\omega$ .  $q = [q_1 \ q_2 \ q_3 \ q_4]^T$  with  $q^T q = 1$  (dimensionless;  $q_4$  is the scalar part);  $\omega$  = body angular velocity (rad/s).

$$R(\mathbf{q}) = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 \\ q_3 & q_4 & -q_1 \\ -q_2 & q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix}$$

- Eq. (2), p. 1347: Euler's equations.  $J$  = inertia matrix ( $\text{kg}\cdot\text{m}^2$ );  $\omega$  = body angular velocity about the principal axes (rad/s);  $\omega^\wedge \times$  = its skew-symmetric matrix;  $M$  = applied torque ( $\text{N}\cdot\text{m}$ ). The estimated  $J$  is printed below the equation.

$$\dot{\mathbf{J}} + \mathbf{J}^T \mathbf{J} = \mathbf{M}$$

$$\mathbf{J} = \begin{bmatrix} 0.00254 & 6.4117 \cdot 10^{-5} & -5.6449 \cdot 10^{-4} \\ 6.4117 \cdot 10^{-5} & 0.00254 & -4.3863 \cdot 10^{-4} \\ -5.6449 \cdot 10^{-4} & -4.3863 \cdot 10^{-4} & 0.0228 \end{bmatrix} \text{ kg}\cdot\text{m}^2$$

- Eq. (3), p. 1348: B-dot control law.  $m_i$  = commanded magnetic dipole on axis  $i$  ( $\text{A}\cdot\text{m}^2$ );  $k_b = 1 \times 10^6$ , a positive gain (units not given);  $\dot{B}_i$  = time derivative of the field on axis  $i$  (units not given).

$$m_i = -k_b \dot{B}_i, \quad i = x, y, z$$

- Eqs. (4)–(5), p. 1348: Scenario 1 settings.  $B$  = augmented field (G;  $1 \text{ G} = 100 \mu\text{T}$ );  $\omega_{\text{init}}$  = initial rate (rad/s).

$$\begin{aligned} \vec{B} &= [-1.670 \quad 0 \quad -0.341]^T \text{ G}, \\ \vec{\omega}_{\text{init}} &= [0 \quad 0 \quad 0.046]^T \text{ rad/s} \end{aligned}$$

### Figures / tables:

- Fig. 1, p. 1346: the z-axis magnetorquer on its 3D-printed frame ( $110 \times 78 \text{ mm}$ ), and the EZHR17EN driver board.
- Table I, p. 1346: magnetorquer parameters. Frame  $110 \times 78 \times 8 \text{ mm}$ ; average winding area  $74.88 \times 10^{-4} \text{ m}^2$ ;  $0.16 \text{ mm}$  wire;  $238.3 \Omega$ ; maximum dipole  $0.248 \text{ A}\cdot\text{m}^2$  at  $50 \text{ mA}$ ;  $50 \text{ g}$ .
- Fig. 2, p. 1347: CubeTAS in the Helmholtz cage, with stereo cameras, sun simulator and air-bearing pedestal; close-up of the floating hemisphere.
- Fig. 3, p. 1347: hardware architecture: ARM single-board computer, IMU, three reaction wheels, coil drivers, sun sensor, battery and DC-DC converter.
- Fig. 4, p. 1347: y- and z-axis magnetorquers fitted to the floating unit.
- Fig. 5, p. 1348: Scenario 1 measured rates. Without control the z-rate stays near  $0.03\text{--}0.04 \text{ rad/s}$ ; with control it reaches about zero within about  $100 \text{ s}$ .
- Fig. 6, p. 1348: Scenario 1 simulation, a near-linear decay to zero at about  $77 \text{ s}$ .
- Fig. 7, p. 1349: Scenario 2, three-axis detumbling with and without control.

### Assumptions / limitations:

- The field was raised to 2.73 times the local field to make the torque visible, so the test doesn't represent the LEO field.

- The cage field is assumed uniform. No map, uniformity figure or cage calibration is reported.
- This is a short correspondence: one run per scenario, with no repeat statistics.
- Disturbances aren't modelled. They're blamed for the 81 s vs 77 s difference.

### Use in my report:

- Section: I-A Problem statement. Claim: "Helmholtz cages are used in CubeSat attitude-control testing, both to recreate the geomagnetic field and to augment it (up to  $\pm 2$  G per axis at NPS)." Use this only if the report frames Wilson's cage as a CubeSat test facility (the same condition as nasa\_smallsats).
- Section: I-A Problem statement, supporting. Claim: "On the ground, field disturbances can be as large as the magnetic control torque, so the test field must be known (p. 1346)."
- Section: IV-G, context only. The local field in the NPS lab was  $[-0.545, -0.07, -0.297]$  G, about  $62 \mu\text{T}$  (p. 1348). That's California, not NZ.

### Unclear / check:

- The torque figures don't match Table I (my arithmetic).  $0.248 \text{ A}\cdot\text{m}^2 \times 170.5 \mu\text{T} = 4.2 \times 10^{-5} \text{ N}\cdot\text{m}$ , but p. 1348 says the maximum is  $6.5 \times 10^{-5} \text{ N}\cdot\text{m}$ . In the unaugmented  $62.5 \mu\text{T}$  field the maximum would be about  $1.5 \times 10^{-5} \text{ N}\cdot\text{m}$ , not the " $\approx 10^{-6} \text{ N}\cdot\text{m}$ " on p. 1348.
- $k_b = 1 \times 10^6$  has no units, and the units of  $\dot{B}$  aren't given.
- $J_{zz} = 0.0228 \text{ kg}\cdot\text{m}^2$  is about nine times  $J_{xx}$  and  $J_{yy}$  (0.00254), which is odd for a nearly symmetric hemisphere. It may be a typo for 0.00228. It doesn't matter for my report.
- "Three to six times larger than LEO" (p. 1348) has no source.
- The number of turns isn't given;  $m = \text{NIA}$  implies about 660 (my inference).
- Gap leads: da Silva et al. 2019 (ref. [5]), Helmholtz cage design and validation. Piergentili et al. 2011 (ref. [15]), closed-loop field intensity and uniformity of a three-axis simulator.

### alken2021international

**International Geomagnetic Reference Field: the thirteenth generation.** P. Alken, E. Thébault, C. D. Beggan, H. Amit, J. Aubert, J. Baerenzung *et al.* (65 authors), *Earth Planets Space*, vol. 73, Art. no. 49, 2021 (25 pp.; published 11 Feb. 2021). DOI 10.1186/s40623-020-01288-x. Cite key provisional.

**Summary (3–4 lines):** In December 2019, IAGA Working Group V-MOD adopted IGRF-13: a definitive model (DGRF) for 2015.0, an IGRF for 2020.0 and a predicted secular variation for 2020–2025. Each final model is the median (or Huber-weighted mean) of 11–14 candidate models from 15 teams, built mainly from Swarm and observatory data. The paper gives the defining equations, the full Gauss-coefficient table (1900–2020, degree  $\leq 13$ ), 2020 maps of declination, inclination and total field, their predicted rates of change, pole positions and the falling dipole moment. It's valid from 1900.0 to 2025.0 and has since been replaced by IGRF-14 (November 2024).

### Key equations:

- Eq. (1), p. 2: scalar potential, with  $B = -\nabla V$ .  $V$  = potential (nT·km);  $r$  = geocentric radial distance (km);  $\theta$  = geocentric colatitude;  $\phi$  = longitude;  $a = 6371.2$  km reference radius;  $g_n^m, h_n^m$  = Gauss coefficients (nT);  $P_n^m$  = Schmidt semi-normalised associated Legendre functions;  $N$  = maximum degree (10 up to 1995, 13 after).

$$V(r, \theta, \phi, t) = a \sum_{n=1}^N \sum_{m=0}^n \left( \frac{a}{r} \right)^{n+1} \left[ g_n^m(t) \cos m\phi + h_n^m(t) \sin m\phi \right] P_n^m(\cos \theta)$$

- Eqs. (2)–(3), p. 2: piecewise-linear time dependence.  $T_t$  = model epoch immediately before  $t$  (years, a multiple of 5, with  $T_t \leq t < T_t + 5$ );  $\dot{g}, \dot{h}$  = secular variation (nT/yr).

$$g_n^m(t) = g_n^m(T_t) + (t - T_t) \dot{g}_n^m(T_t), \quad h_n^m(t) = h_n^m(T_t) + (t - T_t) \dot{h}_n^m(T_t)$$

- Eqs. (4)–(5), p. 14: secular variation between epochs (nT/yr). For 2020–2025,  $\dot{g}$  and  $\dot{h}$  are given directly as a forecast.

$$\dot{g}_n^m(T_t) = \frac{1}{5} \left( g_n^m(T_t+5) - g_n^m(T_t) \right), \quad \dot{h}_n^m(T_t) = \frac{1}{5} \left( h_n^m(T_t+5) - h_n^m(T_t) \right)$$

- Eq. (6), p. 21: dipole moment  $M$  ( $A \cdot m^2$ ), with  $\mu_0$  = vacuum permeability and  $a$  in metres. The axial dipole moment is defined beside it (p. 21, unnumbered).

$$M(t) = \frac{4\pi}{\mu_0} a^3 \sqrt{g_1^0(t)^2 + g_1^1(t)^2 + h_1^1(t)^2}, \quad M_A(t) = \frac{4\pi}{\mu_0} a^3 |g_1^0(t)|$$

### Figures / tables:

- Table 1, p. 2: IGRF generations. IGRF-13 is valid 1900.0–2025.0, definitive 1945.0–2015.0, released 2019.

- Table 2, pp. 3–14: Gauss coefficients (nT) for every epoch 1900–2020, plus the 2020–25 secular variation (nT/yr, degree  $\leq 8$ ).
- Table 3, pp. 16–17: contributing observatories. The NZ row (Institute of Geological and Nuclear Sciences) lists API, EYR and SBA.
- Fig. 1, p. 18: 2020 maps of declination D, inclination I and total field F. F is largest over Siberia and in the Southern Ocean south of Australia, and smallest in the South Atlantic Anomaly. By my reading of the contours, NZ sits in roughly the 55–60  $\mu\text{T}$  band (approximate).
- Fig. 2, p. 19: predicted annual change of D, I and F for 2020–2025.  $dF/dt$  contours reach about  $\pm 100$  nT/yr.
- Fig. 3, p. 20: dip-pole and geomagnetic-pole tracks since 1900.
- Table 4, p. 21: pole positions 1900.0–2025.0 (degrees, WGS84 geodetic).
- Fig. 4, p. 22: dip-pole speeds. The north pole peaked at 55.8 km/yr (2002.5); the 2022.5 forecast is 39.8 km/yr.
- Fig. 5, p. 22: dipole moment falling from about  $8.3 \times 10^{22} \text{ A}\cdot\text{m}^2$  (1900) to about  $7.7 \times 10^{22} \text{ A}\cdot\text{m}^2$  (2025 forecast).

#### **Assumptions / limitations:**

- It models only the large-scale core field (degree  $\leq 13$ ). Crustal anomalies, external (ionospheric and magnetospheric) fields and daily variation aren't included. Indoors, local steel will dominate anyway.
- The secular variation for 2020–2025 is a linear forecast, and past forecasts had significant errors (p. 20).
- Validity ended at 2025.0. A 2026 value needs IGRF-14 (or WMM).

#### **Use in my report:**

- Section: IV-G Field map and map repeatability. Claim: "Earth's main field changes by at most about 100 nT per year (Fig. 2), so the  $\sim 1 \mu\text{T/h}$  drift between maps cannot be geomagnetic secular variation; it points to the sensor or the lab." Daily external variation isn't covered by IGRF, so you need a separate source for that part.
- Section: IV-G, supporting. Claim: "The background geomagnetic field is described by the IGRF spherical-harmonic model (Eq. (1))." Cite this paper for the model definition. Take the actual Wellington value from IGRF-14 or WMM via the NOAA calculator (see `noaa_magcalc`).

- Section: I-A Problem statement, supporting. Claim: "Reference field models such as IGRF are used in satellite attitude determination and control (p. 1), which is why test cages replay model fields along an orbit." This pairs with uscatego2023highprecision.
- It also corrects Sobko's "Earth's field is about 10  $\mu\text{T}$ " (Batch 1). Fig. 1 shows surface F of roughly 25–65  $\mu\text{T}$  worldwide.

#### Unclear / check:

- IGRF-13 expired at 2025.0. IGRF-14 was released on 19 Nov. 2024 (DGRF 2020, IGRF 2025, secular variation 2025–2030). Cite IGRF-14 for any 2025–2026 value.
- The NZ band from Fig. 1 is my contour reading, not a value stated in the paper. Don't quote it; compute the value instead.
- Table 3 lists observatory code API under both New Zealand and Samoa (p. 17). It's minor and not needed.
- Gap leads: the IGRF-14 definition paper in the *Earth Planets Space* special issue "International Geomagnetic Reference Field – The Fourteenth Generation". Its companion is C. D. Beggan *et al.*, "Evaluation of candidate models for the 14th generation International Geomagnetic Reference Field," *Earth Planets Space* 78, 126 (2026), DOI 10.1186/s40623-026-02382-2.

#### noaa\_magcalc

**NCEI Geomagnetic Calculators — Magnetic Field Calculators.** NOAA National Centers for Environmental Information (NCEI), Boulder, CO, USA.

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml> (saved 9 Oct. 2026, 14:13 NZDT, with SingleFile; accessed 9 Oct. 2026). Cite key provisional.

**This file is only the calculator's input form, not a result.** The Declination tab is open, the latitude and longitude boxes are empty, the hemisphere buttons are on their defaults (N and W), and WMMHR with the date 9 Oct. 2026 is preselected. No field value is saved.

**Summary (3–4 lines):** NOAA's online calculator gives declination and the full field vector from the latest World Magnetic Model (WMM), its high-resolution version (WMMHR) or the IGRF. The gufm1 model covers 1590–1900, with a blend into IGRF over 1890–1900. The models offered on the saved page are WMM (2024–2029), IGRF (1590–2029) and WMMHR (2024–2029). The page says declination is typically accurate to 30 arc-minutes, but local magnetic disturbances can be larger. Output can be HTML, XML, CSV, JSON or PDF, and there's an API that needs registration.

**Key equations:** none.

**Figures / tables:** none.

**Assumptions / limitations:**

- The values are model estimates of the main field. Local disturbances (building steel, equipment) and daily variation aren't included.
- The page also mentions the Enhanced Magnetic Model (EMM), a research model that includes finer crustal detail.

**Use in my report:**

- Section: IV-G Field map and map repeatability. Claim, once rerun: "At the lab (latitude, longitude, elevation) on <scan date>, the [IGRF-14/WMM] model gives  $F = \dots \mu\text{T}$ ,  $I = \dots^\circ$ ,  $D = \dots^\circ$  [NOAA calculator, accessed ...]." The coils-off map can then be compared with that value.
- Until it's rerun: Background only.

**Unclear / check:**

- Rerun the calculator on the Magnetic Field tab. Use your lab's coordinates with the S and E hemispheres selected (the saved form defaults to N and W), the scan date and the model. Save the PDF or CSV result, and record which model you used.
- The IGRF option now runs to 2029, so the site is using IGRF-14, not IGRF-13 (see alken2021international).
- Gap leads: none. It's the online tool that alken2021international (p. 21) and uscatego2023highprecision (ref. [17]) point to.

tsekos2007magnetic

**Magnetic Resonance-Compatible Robotic and Mechatronics Systems for Image-Guided Interventions and Rehabilitation: A Review Study.** N. V. Tsekos, A. Khanicheh, E. Christoforou and C. Mavroidis, *Annu. Rev. Biomed. Eng.*, vol. 9, pp. 351–387, 2007 (first online as Review in Advance 17 Apr. 2007). DOI 10.1146/annurev.bioeng.9.121806.160642. Accessed 9 Oct. 2026 (PDF via VUW). Also saved as the Annual Reviews web page, <https://www.annualreviews.org/content/journals/10.1146/annurev.bioeng.9.121806.160642> (same text, no extra content; accessed 9 Oct. 2026, assumed). PDF page = journal page – 350. Cite key provisional.

**Summary (3–4 lines):** A narrative review of robots and mechatronic devices built to work inside MRI scanners, for image-guided interventions (brain, breast, prostate, liver, general purpose) and fMRI/rehabilitation studies. It sets out the MRI hazards (static field, switched gradients, RF), defines MR safety vs MR compatibility, and reviews materials,

actuators (ultrasonic/piezo, pneumatic, hydraulic, ER fluids, Lorentz force), mostly fibre-optic sensors, and compatibility tests. Feasibility is proven; clinical merit, true MR-compatible actuators and stiffer MR-inert materials remain open.

**Key equations:** none (descriptive review).

**Figures / tables:** (no tables)

- Fig. 1, p. 359: commercial ultrasonic motor with controller, and the Johns Hopkins pneumatic stepper (PneuStep, ~0.050 mm steps, tested to 7 T).
- Fig. 2, p. 364: manually actuated prostate manipulator with flexible drive shafts.
- Fig. 3, p. 365: MR images of a prostate needle placement (target and needle void).
- Fig. 4, p. 367: InnoMotion commercial pneumatic robot arm and patient set-up.
- Fig. 5, p. 368: InnoMotion pig study: trajectory planning, tracking markers, placement check.
- Fig. 6, p. 369: Washington University 7-DOF device: Cartesian base plus 4-DOF arm driven through shafts and double U-joints.
- Fig. 7, p. 370: force-sensing wrist module for fMRI.
- Fig. 8, p. 371: pneumatic tactile stimulator.
- Fig. 9, p. 372: 1-DOF haptic interface with hydrostatic transmission; motor outside the scanner room.
- Fig. 10, p. 373: hand-rehabilitation device with rotary ER-fluid brake.
- Fig. 11, p. 374: hand device with linear ER damper.
- Fig. 12, p. 375: electromagnetic haptic interface using the Lorentz force in the scanner field.
- Fig. 13, p. 375: pneumatic parallel-link robotic arm (manipulandum) in a scanner.

**Assumptions / limitations:**

- Compatibility is judged by MR image quality (SNR, artefacts) in 0.3–4 T scanners, not by  $\mu$ T-level magnetometer readings, so material conclusions carry over only by analogy.
- A 2007 snapshot of early prototypes; "only commercial system" (InnoMotion, p. 366) is dated.

**Use in my report:**

- Section: II-B Positioning in magnetic environments. Claim: "Robots for MRI avoid ferromagnetic parts and use plastics, ceramics, fibreglass and composites; a few small metal parts (screws, bearings, gears) are tolerated if small and away from the sensitive volume" (p. 358).

- Section: II-B. Claim: "A common strategy is to keep actuators away from the sensitive region and transmit motion by shafts, belts, cables or linkages" (p. 360). Closest analogy: Chinzei's robot, drive mechanism above the scanner with two long slim members reaching in (p. 363).
- Section: II-D / III-F. Claim: "Remotely actuated joints suffer from joint flexibility, backlash and friction" (p. 360, citing their ref. 98).
- Section: II-D / III-H, supporting. Claim: "Non-metallic structural materials have limited stiffness, which reduces accuracy" (p. 358; Masamune's PET frame too flexible, p. 363).
- Section: IV-E / IV-K, method. Claim: "Following the four-condition test for MR devices (absent; present, unpowered; powered, idle; moving; p. 362), the robot's own magnetic signature at the probe can be checked."
- Section: IV-D, comparison. Masamune's manipulator < 3 mm overall accuracy (p. 363); Krieger's prostate device places needles within 2 mm (p. 364).

#### Unclear / check:

- Error in source: p. 358 groups carbon fibre with "nonmagnetic and nonconductive" materials. Carbon fibre conducts; don't cite this for non-conductivity.
- Terminology: its "MR safe / MR compatible" (p. 357) predates ASTM F2503 (MR Safe / Conditional / Unsafe) used by farooq2023decade (p. 862).
- p. 366 "0.3 open scanner" lacks the unit (0.3 T). Ref. 98's pages are misprinted "2886-71" (p. 386).
- Web page metadata gives 15 Aug. 2007 (volume date) vs online 17 Apr. 2007 in the PDF; cite 2007.
- PDF pp. 1-7 and 26-37 read from the text layer (no figures or equations there); pp. 8-25 read as images.
- Gap leads: [8] Chinzei et al., MICCAI 2000 (drive above scanner; II-B). [86] Chinzei et al., MICCAI 1999 (design criteria; II-B). [88] Schenck 1996, *Med. Phys.* 23:815-50 (material susceptibility; III-C). [98] Christoforou & Tsekos 2006, ICRA (remote-actuated joints; III-F). [99] Koseki et al., MICCAI 2004 (repeatability inside MRI; IV-D, II-E).

#### nzminiaturebearings2026female

**Female Rod End (SI6T/K) M6 x 1.0 Right Hand Thread Bearing.** NZ Miniature Bearings Ltd, SKU 470609610, product web page. <https://nzminiaturebearings.com/product/female-rod-end-si6tk--m6-x-10-right-hand-thread-bearing-sku-470609610.html> (page footer: last updated 06/10/2026; accessed 9 Oct. 2026, assumed). Cite key provisional.

**Summary (3–4 lines):** Supplier listing for the rod ends at the delta robot's parallelogram joints. It gives the designation (SI6T/K, female, right-hand M6 × 1.0), price (NZ\$6.90 + GST each, from Christchurch in 5–10 working days), materials, a dimension list and load ratings. The saved page has no drawing and doesn't define its symbols.

**Key equations:** none.

**Figures / tables:** (spec list, no numbered figures)

- Dimensions: bore  $d = 6$  mm; ball  $dk = 12.7$  mm; inner-ring width  $B = 9$  mm; housing width  $C1_{\max} = 6.75$  mm; housing diameter  $d2_{\max} = 20$  mm; thread  $d3 = M6 \times 1.0$  RH;  $h = 30$  mm (centre to shank end); thread length  $l3_{\min} = 12$  mm; overall length  $L4_{\max} = 40$  mm;  $L5 = 5$  mm; spanner flats  $W = 11$  mm;  $d4 = 10$  mm;  $d5 = 13$  mm;  $r1s_{\min} = 0.3$  mm.
- "a" = 13: tilt angle, unit not given (presumably degrees).
- Load ratings: dynamic 7.2 kN; static 7.65 kN.
- Materials: "Bearing material: GCR15 bearing steel" (ferromagnetic chrome steel); "Material: Aluminium" (presumably the housing).

**Assumptions / limitations:**

- A seller's listing, not a manufacturer datasheet; symbol meanings follow ISO 12240-4 rod-end conventions.
- Load ratings are far above the link loads and say nothing about friction or play.

**Use in my report:**

- Section: III-B Requirements and constraints. Claim: "Each rod end allows a 13° tilt, which limits the joint angles and therefore the workspace."
- Section: III-C Mechanical design. Claim: "The parallelogram joints use SI6T/K female rod ends (M6 × 1.0 RH, 6 mm bore, 12.7 mm ball) at NZ\$6.90 each."
- Section: IV-K Limitations. Claim: "The rod-end balls are GCR15 bearing steel, the only ferromagnetic parts near the probe."
- Section: IV-L Sustainability impact. Cost of low-cost off-the-shelf joints (unit price).

**Unclear / check:**

- Possible conflict with outline: the outline says only the balls are steel. The listing says aluminium housing, but SI..T/K housings are often steel. Magnet-test a rod end and correct IV-K if needed.
- "a: 13" has no unit;  $h$ ,  $l3$ ,  $L4$ ,  $L5$ ,  $W$ ,  $d4$ ,  $d5$  are undefined. Check against a drawing before quoting.

- The "T/K" suffix (often a PTFE-lined, maintenance-free type) isn't explained.
- Page glitches "6. 75 mm" and "7. 2 kN" read as 6.75 mm and 7.2 kN. Price and stock change; quote with the access date.
- Gap leads: ISO 12240-4 (rod-end dimensions and tilt angle); the SI6T/K manufacturer datasheet (housing material, liner, tilt angle).

omelyanchik2023doityourself

### A do-it-yourself approach for developing a magnetic field mapping setup using a 3D printer.

A. Omelyanchik, J. L. Marqués, M. Rivas, V. Rodionova, F. Canepa and D. Peddis, *Meas. Sci. Technol.*, vol. 34, Art. no. 107001, 2023 (Technical Note, 9 pp.; published 30 Jun. 2023). DOI 10.1088/1361-6501/acde9b. Open access (CC BY 4.0). Accessed 9 Oct. 2026. PDF page = journal page + 1 (PDF p. 1 is the IOP cover). Cite key provisional.

**Summary (3–4 lines):** The authors replaced the extruder of a ~€100 Anet A8 3D printer with a printed Hall-probe holder, drove it by G-code from LabVIEW and read a Lake Shore 475 gaussmeter at each grid point. They tuned the settling delay and step size on an N35 test magnet, then mapped a smartphone, a fridge magnet (3.2 mm Halbach period) and a floppy disk at 0.1 mm steps, claiming sub-millimetre resolution and sensitivity below 1 G. They say commercial 3D mappers cost over €25,000 and a Python/Arduino/cheap-sensor version could cost under €200.

### Key equations:

- Unnumbered, p. 2: Hall-element offset.  $z$  = position of the probe-stem surface (mm);  $z_{\text{active}}$  = centre of the 1 mm sensing area inside the stem (mm).

$$z_{\text{active}} = z + 1.9 \text{ mm}$$

- Unnumbered, p. 5: end point of each arrow in the 3D vector plots.  $(x, y, z)$  = measurement point (mm);  $B_x, B_y, B_z$  = flux-density components (G);  $k$  = scale factor from gauss to plotted mm (G/mm).

$$\left(x_{\text{end}}, y_{\text{end}}, z_{\text{end}}\right) = \left(x + \frac{B_x}{k}, y + \frac{B_y}{k}, z + \frac{B_z}{k}\right)$$

### Figures / tables: (no tables)

- Fig. 1, p. 3: (a) workflow PC → G-code → printer → gaussmeter → PC; (b) photo of the set-up; (c) test magnet under the probe.
- Fig. 2, p. 3: LabVIEW front panel: settings, jog control, scan sequence, live 3D plot.

- Fig. 3, p. 4: axial, transverse and 3-axis Hall probes and the B component each measures.
- Fig. 4, p. 5: (a)  $B_x$ ,  $B_y$ ,  $B_z$  maps of the test magnet at  $z = 5.9$  mm for  $t_d = 0.1, 1, 5, 10$  s; (b)  $\sigma$  (%) vs  $t_d$ : highest ( $\sim 3\%$  for  $B_x$ ) at 0.1 s, near zero from 5 s.
- Fig. 5, p. 5:  $B_z$  map at 5, 2 and 1 mm steps (7, 40 and 170 min).
- Fig. 6, p. 6: 3D vector plots at three heights (caption  $z = 1.9, 5.9, 9.9$  mm; panel titles 0, 4, 8 mm).
- Fig. 7, p. 6: log-scale  $|B_z|$  over a smartphone (speaker, camera, vibration motor).
- Fig. 8, p. 7: fridge-magnet  $B_z$  map and 1D profile with sinusoidal fit (period  $3.2 \pm 0.1$  mm, amplitude  $204 \pm 5$  G), Halbach sketch.
- Fig. 9, p. 7:  $B_z$  over a floppy disk ( $\pm 0.3$  G scale) after subtracting the ambient field.

### Assumptions / limitations:

- Position accuracy (0.012 mm x/y, 0.004 mm z) is the printer maker's claim; probe position error was never measured.
- Fields are mT-T near magnets. Below  $\sim 1$  G (100  $\mu$ T) they say the positioner's steel parts must be replaced with non-magnetic ones and ambient fields isolated (p. 7).
- Vectors were built by swapping and re-orienting single-axis probes, so components weren't measured at the same time.
- " $<€200$ " is a projection; the built system used a lab gaussmeter and LabVIEW. Data available only on request.

### Use in my report:

- Section: I-A Problem statement. Claim: "Commercial 3D field mappers cost more than  $€25,000$ " (p. 2, citing their refs 8–10).
- Section: II-A Magnetic field mapping systems. Claim: "Low-cost DIY mappers repurpose 3D printers as Cartesian probe positioners, with sub-millimetre resolution near permanent magnets" (pp. 2, 6–7).
- Section: II-A / I-A, supporting. Claim: "To measure fields below about 1 G with a printer-based mapper, its steel parts must be replaced with non-magnetic materials" (p. 7).
- Section: II-E How this project differs. Claim: "Earlier DIY mappers rely on the positioner's quoted accuracy rather than measuring position error in situ" (pp. 2, 4).
- Section: III-G Scan software. Claim: "A settling delay is needed after each move; scatter fell to near zero with a 5 s delay, averaging  $N = 10$  readings at 100 ms" (Fig. 4, pp. 4–5). Also: finer steps cost much more time,  $7 \rightarrow 40 \rightarrow 170$  min for  $5 \rightarrow 2 \rightarrow 1$  mm (Fig. 5).

- Section: IV-L / I-D (SDG 4). Claim: "DIY mappers are useful teaching tools in physics and engineering" (pp. 7–8).

#### Unclear / check:

- Text vs figure: p. 4 says  $\sigma$  "increases to its maximum" of 3 % for  $t_d \leq 5$  s, but Fig. 4(b) shows  $\sigma$  largest at the shortest delay. Quote the figure.
- p. 4 "N = 10 s" should be N = 10 readings. p. 7 "lower resolution" means finer resolution.
- Fig. 6 panel heights (0/4/8 mm) vs caption (1.9/5.9/9.9 mm) differ by the 1.9 mm sensor offset; the text's "4 mm from the surface" (p. 4) is  $z = 5.9$  mm.
- Accuracy is 0.012 mm on p. 2 but ">0.012 mm" on p. 4; both are manufacturer figures. Don't cite "<€200" as achieved.
- Its refs [9] GMW MMS-1A-RS and [10] SENIS mappers are the senis\_mms product (Batch 1).
- Gap leads: [8] Erglis, Cimurs & Kitenbergs, "Low-cost solution for magnetic field mapping device," 22nd Int. Conf. Electronics (IEEE), 2018, pp. 1–5 (II-A). [13] Vavoulas et al., "Using a modified 3D-printer for mapping the magnetic field of RF coils...", *J. Magn. Reson.* 269, 146–51, 2016 (II-A). [14] Han et al., "Open source 3D multipurpose measurement system with submillimetre fidelity...", *Sci. Rep.* 7, 2017 (II-A).

#### farooq2023decade

**A Decade of MRI Compatible Robots: Systematic Review.** M. U. Farooq and S. Y. Ko, *IEEE Trans. Robot.*, vol. 39, no. 2, pp. 862–884, Apr. 2023 (online 25 Oct. 2022). DOI 10.1109/TRO.2022.3212626. Accessed 9 Oct. 2026 (VUW, IEEE Xplore). PDF page = journal page – 861. Cite key provisional.

**Summary (3–4 lines):** A systematic review (Scopus + PubMed) of MRI robots, actuators and sensors reported after Elhawary's 2008 review. It covers pneumatic, piezoelectric, MR-powered, hydraulic and SMA/optical actuators; fibre-optic force, tactile and shape sensors; fMRI-study robots; and surgical robots by organ, with tables of DOF, actuation, accuracy and SNR drop. Ultrasonic/piezo (36 %) and pneumatic (34 %) actuation dominate; remote drives via cables or shafts suffer hysteresis and friction; few systems reached human trials.

**Key equations:** none (descriptive review).

#### Figures / tables:

- Table I, p. 863: earlier review articles (2008–2020) and their topics.

- Fig. 1, p. 864: classification of the included literature (actuators, sensors, fMRI robots, surgical robots).
- Fig. 2, p. 865: timeline of MRI-compatible actuators, 2011–2020.
- Table II, p. 865: pneumatic motors compared: mechanism, size, torque, speed, SNR drop, pros/cons.
- Fig. 3, p. 867: timeline of MRI-compatible sensors, 2009–2021.
- Table III, p. 868: force sensors compared: principle, DOF, range, resolution, error, sensitivity.
- Fig. 4, p. 869: timeline of fMRI and surgical robots, 2008–2021, human-trial systems highlighted (includes the delta-like Elayaperumal robot [144]).
- Table IV, p. 873: prostate robots: DOF, actuation, accuracy (mm), status, SNR drop.
- Table V, p. 874: breast robots, same columns.
- Table VI, p. 875: neurosurgery robots, same columns.

#### **Assumptions / limitations:**

- Medical robots only; commercial systems and patents excluded; inclusion depends on the authors' definition of "robot".
- "Accuracy" mixes positioning, targeting and needle-placement errors from phantom, animal and human tests; not comparable with ISO 9283.

#### **Use in my report:**

- Section: II-B Positioning in magnetic environments. Claim: "In MRI robotics, ultrasonic/piezoelectric (36 %) and pneumatic (34 %) actuators dominate, followed by hydraulic (11 %) and manual (11 %)" (p. 878).
- Section: II-B / III-F. Claim: "Low-cost motors placed outside the field, with torque sent by flexible shafts, tendons or Bowden cables, are common, but hysteresis and friction are the main problems" (p. 878).
- Section: II-B. Claim: "Safety terms follow ASTM F2503 (MR Safe, MR Conditional, MR Unsafe); 'MRI compatible' is not an FDA term" (p. 862). Also: off-the-shelf bearings, gears and belts must be chosen carefully (p. 864).
- Section: II-C Delta robots, lead only. Claim: "Delta-like parallel mechanisms have been used in MRI-guided devices" (p. 872, ref. [144]); read the primary paper first.
- Section: III-F Firmware. Claim: "Open-loop MRI robots handle joint backlash by compensating for it in the kinematics" (p. 874, Price [193]); Wang et al. [183] report open-loop control with  $\pm 1.66$  mm backlash (p. 874).

- Section: IV-D, comparison. InnoMotion precision  $\pm 1$  mm,  $\pm 1^\circ$  (p. 877); Stormram 3 accuracy 2 mm (p. 873); low-cost pneumatic steppers with  $\sim 2^\circ$  maximum angular error (p. 864).

### Unclear / check:

- The abstract says 2009–2021 (p. 862); the search in §I-F covered 2008–2021 (p. 863).
- Table IV (p. 873) gives ref. [160] to both "Jiang et al." and "Giannakou et al.", and lists Yiallouras [156] as "20  $\mu\text{m}$ " in a mm column.
- Table V (p. 874) gives Stormram 4 [172] 0.7 mm accuracy; the text (p. 873) gives  $1.29 \pm 0.59$  mm.
- Terminology differs from tsekos2007magnetic (MR safe / compatible); pick ASTM F2503 terms.
- PDF pp. 9–10 (pp. 870–871) and 18–23 (references) read from the text layer; no tables or equations there.
- Gap leads: [144] Elayaperumal et al., "A passive parallel master–slave mechanism for MRI-guided interventions," *J. Med. Devices* 9(1), 2015 (II-B, II-C). [183] Wang, Cheng & Desai, "Design, analysis, and evaluation of a remotely actuated MRI-compatible neurosurgical robot," *IEEE RA-L* 3(3), 2144–2151, 2018 (II-B, IV-D). [193] Price et al., "Design and validation of an MR-conditional robot for transcranial focused ultrasound surgery in infants," *Med. Phys.* 43(9), 4983–4995, 2016 (III-F). [14] Elhawary et al., "The case for MR-compatible robotics," *Int. J. Med. Robot.* 4(2), 105–113, 2008 (II-B). [25] Xiao et al., "MR-conditional actuations: a review," *Ann. Biomed. Eng.* 48(12), 2707–2733, 2020 (II-B).

### yang2020analytical

#### Analytical elastostatic stiffness modeling of parallel manipulators considering the compliance of the link and joint.

C. Yang, Q. Li and Q. Chen, *Applied Mathematical Modelling*, vol. 78, pp. 322–349, Feb. 2020 (received 18 Mar. 2019; online 17 Oct. 2019). doi: 10.1016/j.apm.2019.10.024. Web copy:

<https://www.sciencedirect.com/science/article/pii/S0307904X19306122> (via VUW Library, Open archive; accessed 9 Oct. 2026). Text-based PDF, not scanned. Cite key provisional.

**Web copy (same article).** The batch's saved ScienceDirect page (S0307904X19306122) is this article. Abstract, conclusion, Section 3.5 text, Table 5 values and  $\zeta = 0.635$  match the PDF. It adds four Highlights and a Scopus "Cited by (25)" list. Its equations and tables are images, so it was not used to cross-check the LaTeX. One Zotero item with both attachments.

**Summary (3–4 lines):** A closed-form stiffness model for parallel manipulators (PMs) that includes link and joint compliance. Screw theory gives each limb's constraint wrenches; strain energy and Castigliano's second theorem give a compact limb compliance matrix; deformation compatibility assembles the overall stiffness. Joint compliances (R-, cross U-, T-shape U-, S-joints) come from matrix structural analysis (MSA) with Timoshenko beams. On a 2PRU-UPR example the model matches ANSYS FEA within 8.87 %; ignoring joint flexibility raises the error to 39.84 %. A strain energy factor index (SEFI) ranks the weakest component.

**Key equations** (units: E, G in Pa; A in m<sup>2</sup>; I in m<sup>4</sup>; L, s in m; forces N; couples N·m; deflections m and rad; U in J):

- Eq. (1), p. 326: total link strain energy of limb i;  $U_i^B = \sum_j U_{ij}^B$  = strain energy of link j in limb i.

$$U_i^B = \sum_j U_{ij}^B$$

- Eq. (2), p. 326: link deflection along the constraint wrenches (Castigliano).  $C_i^B = a_i \times a_i$  link compliance (layout in Eq. (3), p. 327);  $w_i = a_i \times 1$  constraint-wrench magnitudes ( $c_i$  forces,  $t_i$  couples).

$$\boldsymbol{\delta}_i^B = \mathbf{C}_i^B \mathbf{w}_i$$

- Eq. (4), p. 327: link strain energy as a quadratic form.

$$U_i^B = \frac{1}{2} \mathbf{w}_i^T \boldsymbol{\delta}_i^B = \frac{1}{2} \mathbf{w}_i^T \mathbf{C}_i^B \mathbf{w}_i$$

- Eq. (5), p. 327: wrenches translated to joint frame  $\{o_{ij}\}$ ;  $D_{ij} = 6 \times a_i$  mapping matrix.

$$\mathbf{w}_{ij}^{o_{ij}} = \mathbf{D}_{ij} \mathbf{w}_i$$

- Eq. (6), p. 327: strain energy of joint j in limb i;  $C_{ij}^J = 6 \times 6$  joint compliance in joint space (supplier, FEA, or Section 2.3).

$$U_{ij}^{J_i} = \frac{1}{2} \mathbf{w}_{ij}^{o_{ij}T} \boldsymbol{\delta}_{ij}^{o_{ij}} = \frac{1}{2} \mathbf{w}_{ij}^{o_{ij}T} \tilde{\mathbf{C}}_{ij}^J \mathbf{w}_{ij}^{o_{ij}},$$

$$\mathbf{w}_{ij}^{o_{ij}} = \frac{1}{2} \mathbf{w}_i^T \mathbf{C}_{ij}^J \mathbf{w}_i$$

- Eq. (7), p. 327: joint compliance mapped into constraint-wrench space ( $a_i \times a_i$ ).

$$\mathbf{C}_{ij}^J = \mathbf{D}_{ij}^T \tilde{\mathbf{C}}_{ij}^J \mathbf{D}_{ij}$$

- Eq. (8), p. 327: strain energy of all joints in limb  $i$  (source prints the last factor as  $w^i$ ; should be  $w_i$ ).

$$U_i^J = \sum_j U_{ij}^J = \frac{1}{2} \mathbf{w}_i^T \mathbf{C}_i \mathbf{w}_i, \quad \mathbf{C}_i = \mathbf{C}_i^B + \sum_j \mathbf{D}_{ij}^T \tilde{\mathbf{C}}_{ij}^J \mathbf{D}_{ij}$$

- Eqs. (9)–(10), p. 327: limb strain energy and limb compliance (links in series with joints).

$$U_i = U_i^B + U_i^J = \frac{1}{2} \mathbf{w}_i^T \mathbf{C}_i \mathbf{w}_i, \quad \mathbf{C}_i = \mathbf{C}_i^B + \sum_j \mathbf{D}_{ij}^T \tilde{\mathbf{C}}_{ij}^J \mathbf{D}_{ij}$$

- Eqs. (11)–(12), p. 328: limb stiffness and limb deflection.

$$\mathbf{K}_i = \mathbf{C}_i^{-1} = \left( \mathbf{C}_i^B + \sum_j \mathbf{D}_{ij}^T \tilde{\mathbf{C}}_{ij}^J \mathbf{D}_{ij} \right)^{-1}, \quad \boldsymbol{\delta}_i = \mathbf{C}_i \mathbf{w}_i$$

- Eq. (13), p. 328: Euler–Bernoulli cantilever stiffness (no shear; the authors say this fails for short beams).

$$\tilde{\mathbf{K}} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{12EI_z}{L^3} & 0 & 0 & -\frac{6EI_z}{L^2} & 0 \\ 0 & 0 & \frac{12EI_y}{L^3} & 0 & -\frac{6EI_y}{L^2} & 0 \\ 0 & 0 & 0 & \frac{GI_p}{L} & 0 & 0 \\ 0 & -\frac{6EI_z}{L^2} & 0 & 0 & \frac{6EI_z}{L} & 0 \\ 0 & -\frac{6EI_y}{L^2} & 0 & 0 & 0 & \frac{6EI_y}{L} \end{bmatrix}$$

- Eq. (14), p. 328: section forces and couples at distance  $s_c$  from the load point, under tip wrench  $\mathbf{w}_c = [F_{xc}, F_{yc}, F_{zc}, M_{xc}, M_{yc}, M_{zc}]^T$ .

$$f_{xc} = F_{xc}, \quad f_{yc} = F_{yc}, \quad f_{zc} = F_{zc}; \quad m_{xc} = M_{xc}, \quad m_{yc} = -s_c F_{zc} + M_{yc}, \quad m_{zc} = s_c F_{yc} + M_{zc}$$

- Eq. (15), p. 329: Timoshenko cantilever compliance;  $A_y, A_z$  = shear areas.

```
\mathbf{C}_{ec}=\begin{bmatrix} \frac{L}{EA}&0&0&0&0&0\\ 0&\frac{L}{GA_y}+\frac{L^3}{3EI_z}&0&0&0&0\\ \frac{L^2}{2EI_z}&0&\frac{L}{GA_z}+\frac{L^3}{3EI_y}&0&0&0\\ 0&0&0&\frac{L}{GI_p}&0&0\\ 0&0&0&0&\frac{L^2}{2EI_y}&0\\ 0&0&0&0&0&\frac{L}{EI_z} \end{bmatrix}
```

- Eq. (16), p. 329: Timoshenko cantilever stiffness (symmetric; upper triangle shown).

```
\mathbf{K}_{ec}=\begin{bmatrix} \frac{EA}{L}&0&0&0&0&0\\ &\frac{12EI_{zA_y}}{Lg_1}&0&0&0&0\\ &-\frac{6EI_{zA_y}}{Lg_1}&\frac{12EI_{yA_z}}{Lg_2}&0&0&0\\ &0&\frac{6EI_{yA_z}}{Lg_2}&\frac{GI_p}{L}&0&0\\ &0&0&0&\frac{4EI_{yb_1}}{Lg_2}&0\\ &0&0&0&0&\frac{4EI_{zb_2}}{Lg_1} \end{bmatrix}, \quad g_1=GA_yL^2+12EI_z, \quad g_2=GA_zL^2+12EI_y, \\ b_1=GA_zL^2+3EI_y, \quad b_2=GA_yL^2+3EI_z
```

- Eqs. (17)–(28), pp. 329–331: unsupported-beam element  $K_{leu}$  (blocks  $K_{l11}$ ,  $K_{l12}$ ,  $K_{l22}$ ) and assembly of the  $18 \times 18$  extended matrix for the R-joint shaft (two elements, three nodes),  $K_e = K_{e1} + K_{e2}$ . Standard FE bookkeeping; not transcribed.
- Eqs. (29)–(30), p. 331: R-joint stiffness = middle-node block of  $K_e$  (circular section:  $g = g_1 = g_2$ ,  $b = b_1 = b_2$ ); the zero is the free rotation about the shaft. Last two terms printed with  $A_y$ ,  $A_z$ ; by dimensions they should be  $I_y$ ,  $I_z$  (see Flags).

```
\mathbf{K}_R=\mathbf{K}_e(7\!:\!12,7\!:\!12)=\operatorname{diag}\Big(\tfrac{2EA}{L},\tfrac{24EGA_{yI_z}}{Lg},\tfrac{24EGA_{zI_y}}{Lg},0,\tfrac{8EA_{yb}}{Lg},\tfrac{8EA_{zb}}{Lg}\Big) \quad \text{(as printed)}, \quad \mathbf{C}_R=\mathbf{K}_R^{-1}
```

- Eqs. (31)–(36), p. 332: cross U-joint as four elements, five nodes ( $30 \times 30$ ); elements rotated to the global frame by  $K_{geu,i} = B_i^T K_{leu,i} B_i$  and summed. Not transcribed.
- Eqs. (37)–(38), p. 333: cross U-joint stiffness (two rotations free; same A-vs-I issue in the last term).

```
\mathbf{K}_{Uc}=\operatorname{diag}\Big(\tfrac{2EA}{L}+\tfrac{24EGA_{yI_z}}{Lg},\tfrac{2EA}{L}+\tfrac{24EGA_{yI_z}}{Lg},\tfrac{48EGA_{zI_y}}{Lg},0,0,\tfrac{16EA_{zb}}{Lg}\Big), \quad \mathbf{C}_{Uc}=\mathbf{K}_{Uc}^{-1}
```

- Eq. (39), p. 333: T-shape U-joint stiffness (derivation omitted in the paper); diagonal below, plus symmetric  $K_{UT}(1,6) = K_{UT}(6,1) = 6EGA_{yl_z}/g$ .

$$\operatorname{diag}(\mathbf{K}_{UT}) = \operatorname{Big}(\frac{2EA}{L} + \frac{12EGA_y I_z}{Lg}, \frac{2EA}{L} + \frac{24EGA_y I_z}{Lg}, \frac{36EGA_z I_y}{Lg}, 0, 0, \frac{12EA_z b}{Lg})$$

- Eqs. (40)–(41), p. 334: S-joint as three orthogonal shafts; translational stiffness only. Shaft radius  $r_s = \zeta r$ , with  $\zeta = 0.635$  fitted in ANSYS for a steel sphere ( $E = 210$  GPa,  $G = 80$  GPa).

$$\mathbf{K}_S = \operatorname{diag}(\frac{2EA}{L} + \frac{48EGA_y I_z}{Lg}, \frac{2EA}{L} + \frac{48EGA_y I_z}{Lg}, \frac{2EA}{L} + \frac{48EGA_y I_z}{Lg}, 0, 0, 0) \quad r_s = \varsigma r, \quad \varsigma = 0.635$$

- Eq. (42), p. 335: platform equilibrium.  $\mathbf{p} = [\mathbf{f}^T; \mathbf{m}^T]^T$  = external wrench at point o;  $\mathbf{J}_i = [\mathbf{r}_{i1}, \mathbf{r}_{i2}, \dots] = 6 \times \mathbf{a}_i$  matrix of unit constraint screws (not a velocity Jacobian).

$$\mathbf{p} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{J}_2 & \cdots & \mathbf{J}_n \end{bmatrix} \mathbf{w} = \mathbf{J} \mathbf{w}$$

- Eqs. (43)–(44), p. 335: virtual work for a rigid platform gives deformation compatibility;  $\Delta$  = infinitesimal twist of the platform at o.

$$\mathbf{p}^T \boldsymbol{\Delta} = \sum_{i=1}^n \mathbf{w}_i^T \boldsymbol{\delta}_i, \quad \boldsymbol{\delta}_i = \mathbf{J}_i^T \boldsymbol{\Delta} \quad (i=1, \dots, n)$$

- Eqs. (45)–(47), p. 335: overall stiffness and compliance (main result).

$$\mathbf{p} = \sum_{i=1}^n \mathbf{J}_i \mathbf{K}_i \mathbf{J}_i^T \boldsymbol{\Delta} = \mathbf{K} \boldsymbol{\Delta}, \quad \mathbf{K} = \sum_{i=1}^n \mathbf{J}_i \mathbf{K}_i \mathbf{J}_i^T, \quad \mathbf{C} = \mathbf{K}^{-1}$$

- Eqs. (48)–(49), p. 335: constraint-wrench magnitudes and total strain energy for a given load.

$$\mathbf{w}_i = \mathbf{K}_i \mathbf{J}_i^T \mathbf{C} \mathbf{p}, \quad U = \sum_{i=1}^n U_i = \frac{1}{2} \mathbf{p}^T \mathbf{C} \mathbf{p}$$

- Eqs. (50)–(51), p. 336: local SEFI (0 to 1) and global SEFI averaged over the workspace;  $N$  = number of discrete workspace points.

$$\mu_{ij} = \frac{U_{ij}}{U}, \quad \bar{\mu}_{ij} = \frac{\int_V \mu_{ij} dV}{\int_V dV} = \frac{1}{N} \sum \mu_{ij}$$

- Eqs. (52)–(72), pp. 337–339: twist and constraint screws, section forces, link compliances  $C^A B_i$  (Eqs. (61)–(62), with cross-section factor  $\lambda_i$ ) and the D matrices for the 2PRU-UPR. Example-specific; not transcribed.
- Eq. (73), p. 342: 2PRU-UPR reachable-workspace limits.

$$50 \text{ mm} \leq q_1, q_2 \leq 1200 \text{ mm}; \quad 100 \text{ mm} \leq q_3 \leq 1000 \text{ mm}; \quad |\alpha_{i1}|_{\max} \leq 50^\circ, \quad |\alpha_{i2}|_{\max} \leq 50^\circ$$

### Figures / tables:

- Fig. 1, p. 325: general PM with limbs, passive and active joints, frames.
- Fig. 2, p. 326: three-module flowchart (wrenches → limb compliance → overall stiffness).
- Fig. 3, p. 327: one limb with its joint and link frames.
- Fig. 4, p. 328: cantilever and unsupported beam elements.
- Fig. 5, p. 330: R-joint schematic, FE model, overall force and free-body diagrams.
- Figs. 6–7, pp. 332–333: cross U-joint and T-shape U-joint schematics and FE models.
- Figs. 8–9, p. 334: S-joint as three shafts; CAD model and ANSYS mesh used to fit  $\zeta$ .
- Figs. 10–11, pp. 336–337: 2PRU-UPR CAD model and screw schematic; PRU and UPR free-body diagrams.
- Fig. 12, p. 341: FEA deformation contours; max about  $3.5 \times 10^{-5}$  m (configuration 1) and  $4.6 \times 10^{-5}$  m (configuration 2).
- Fig. 13, p. 342: reachable workspace and the elliptical-cylinder regular workspace.
- Figs. 14–16, pp. 343–344: total strain energy and per-component SEFI over the workspace and in the  $z = 600$  mm plane.
- Figs. 17–20, pp. 345–346: constraint-wrench magnitudes (cases 1–2) and how U- and R-joint  $\mu$  splits by wrench component.
- Figs. 21–22, pp. 346–347: link SEFI slices under two other wrenches.
- Fig. 23, p. 347: MATLAB scatter and correlation matrices of SEFI vs  $z$ ,  $\beta$ ,  $\gamma$  for three wrenches.

- Table 1, p. 335: S-joint diagonal term, FEA vs theory,  $R = 10\text{--}50\text{ mm}$ ; errors  $\leq 0.6\%$  (rechecked).
- Table 2, p. 340: numeric  $6 \times 6$  compliance matrices of the R-, cross U- and T-shape U-joints ( $\times 10^{-10}$ , N/m/rad);  $\infty$  = free DOF.
- Table 3, p. 340: cross U-joint stiffness, FEA vs Euler–Bernoulli MSA (errors up to 102.4 %) vs Timoshenko MSA ( $\leq 2.7\%$ ); rechecked, consistent with Table 2.
- Table 4, p. 341: platform deflection, analytical vs FEA, two configurations; max error 8.87 % (rechecked).
- Table 5, p. 342: four scenarios; with rigid joints the max error is 39.84 % (config 1) and 23.70 % (config 2).
- Table 6, p. 343: global SEFI per component, three wrenches; under a pure z-force the links hold 68.3 % of the strain energy, the joints 31.7 % (rechecked).

#### Assumptions / limitations:

- Component weights neglected, friction ignored, platform, base and actuators rigid (pp. 324–325).
- Covers PMs with simple sub-closed loops such as a Delta's parallelograms (their ref. [26]); complicated sub-closed loops left to future work (p. 325).
- Validated only against FEA; no experiment.
- Example is a large steel 2PRU-UPR (60 mm links,  $E = 210\text{ GPa}$ ); magnitudes don't carry over to a PLA/carbon-fibre delta.
- Results are local to a configuration; global SEFI needs a workspace sweep.  $\zeta = 0.635$  fitted for steel only.

#### Use in my report:

- Section: II-D. Claim: "In parallel manipulators, joint and link compliance (non-geometric error) is significant: in Yang et al.'s example, ignoring joint flexibility raised the deflection error from 8.87 % to 39.84 % against FEA" (Tables 4–5, pp. 341–342).
- Section: III-H, framework only. Claim: "Platform deflection under load can be found by mapping each limb's compliance to the platform through Jacobian-type matrices,  $K = \sum J_i K_i J_i^T$  and  $\Delta = C p$ " (Eqs. (45)–(47), p. 335). See Flags: Yang assumes rigid actuators.
- Section: IV-J, supporting. Claim: "Elastic error depends on configuration and load direction" (strain energy and SEFI vary across the workspace, Figs. 14–16, pp. 343–344).

#### Unclear / check:

- Eqs. (29), (37), (39): rotational terms printed with  $A_y$ ,  $A_z$  give N/m, not N·m/rad; probably  $I_y$ ,  $I_z$ .
- Eq. (19):  $K_{12}$  marked "sym.", but a beam's off-diagonal block isn't symmetric (6EI couplings change sign); the rotational entries check out algebraically.
- Table 1: columns labelled "(mm)" with values around  $10^{-7}$ , so they look like compliance; text says stiffness; "R" vs "r" in Eq. (41).
- Conclusion (p. 348) says  $< 8\%$  vs FEA; Table 4 shows 8.87 % and Section 3.5 says  $< 9\%$ .
- Table 5, config 1, scenario 3:  $\Delta I_z = +0.010$  (all others negative); config 1, scenario 2  $\Delta I_x$  error recomputes to 4.19 %, not 4.23 %.
- Small typos: Eq. (8)  $w^i$ ; nomenclature " $\delta_B$ ", " $UB$ "; Figs. 17–18 captions "(a)" for (c); Fig. 13 legend "regualr".

futaba2010s3306mg

**S3306MG: Servo for 1/5 scale cars — high torque, dual ball bearing, metal gear.** Futaba Corporation, product datasheet (1 p.), dated Dec. 2010 ("2010.12"). URL and access date not in the file; add where it was downloaded. Cite key provisional.

**Summary (3–4 lines):** One-page spec sheet for a high-torque, dual-ball-bearing, metal-gear hobby servo for 1/5-scale cars. All ratings are at 6.0 V. No voltage range, pulse range, deadband, current or backlash figures.

**Key equations:** none in the source. My unit checks (not in the datasheet):

$$\tau = 24.0 \text{ kgf}\cdot\text{cm} = 24.0 \times 9.80665 \times 0.01 \approx 2.35 \text{ N}\cdot\text{m} \quad (\text{matches } 333 \text{ oz}\cdot\text{in} \approx 2.35 \text{ N}\cdot\text{m})$$

$$\omega = \frac{60^\circ}{0.16 \text{ s}} = 375^\circ/\text{s} \approx 6.5 \text{ rad/s} \quad (\text{at } 6.0 \text{ V, presumably no load})$$

$$F_{\text{elbow}} \approx \frac{2.35 \text{ N}\cdot\text{m}}{0.177 \text{ m}} \approx 13 \text{ N} \quad (L_1 = 177 \text{ mm})$$

**Figures / tables:**

- Spec box, p. 1: speed 0.16 s/60° at 6.0 V; torque 24.0 kg·cm (333 oz·in) at 6.0 V; size 66 × 30 × 57.1 mm (2.60 × 1.18 × 2.25 in); weight 130 g (4.59 oz). Imperial conversions consistent.

**Assumptions / limitations:**

- Ratings hold only at 6.0 V; not stated whether the torque is stall torque.
- No electrical, pulse-width or deadband data; gear material only "metal".

### Use in my report:

- Section: II-G. Claim: "The servos are Futaba S3306MG, rated 24.0 kg·cm ( $\approx 2.35$  N·m) and 0.16 s/60° at 6.0 V, 130 g each."
- Section: III-C. Claim: "Each servo gives about 2.35 N·m at the bicep (torque basis for arm sizing); the servos have metal gears, part of why they sit more than 600 mm above the probe."
- Section: III-D, supporting. Claim: "The 6 V pack matches the servo's rated 6.0 V."

### Unclear / check:

- No URL or download source in the PDF.
- Your 800–2200  $\mu$ s travel (working  $\sim$ 830–2170  $\mu$ s), 1520  $\mu$ s neutral and  $\sim$ 4° output play are your measurements, not datasheet values.
- No deadband, backlash or hysteresis figures for IV-B.
- 4  $\times$  AA pack sags under load, so real torque and speed may fall below the 6.0 V rating.

## Flags

Cross-paper issues: wrong values, unsupported claims, missing sources. Add each as a task so it can be ticked off.

- ☐ **The ">€25,000" claim doesn't hold for every commercial system** (metrolab\_hallinsight, senis\_mms). The HallinSight 32 $\times$ 2 line camera is US\$17,950. Either limit the claim to 3D mappers (the Senis MMS-1A-RS is US\$143,180) or give a range, e.g. "from about US\$18,000 for a line camera to over US\$140,000 for a 3D mapper."
- ☐ **Wang's formulas are for circular coils, but Wilson's cage has square coils** (wang2002improved). Find a square-coil source before the comparison with Helmholtz theory.
- ☐ **Sobko has two errors not to repeat** (sobko2024development). It gives Earth's field as about 10  $\mu$ T, and it calls the HMC5883 a Hall sensor when it's actually AMR (magnetoresistive).
- ☐ **The CubeSat page is background only** (nasa\_smallsats) unless the report says Wilson's cage is for CubeSat testing.

- ☐ **No source yet for "commercial mappers put motors and steel next to the probe".** Neither vendor page says where the motors sit.
- ☐ **Cite keys are provisional.** Swap them for the keys in your .bib once it's uploaded.
- ☐ Verify wang2002improved Eqs. (1)–(4) and Table I (pp. 2175 and 2178) against the PDF.
- ☐ Verify sobko2024development Eqs. (1)–(3) (p. 287) and Table 1 (pp. 285–286) against the PDF.
- ☐ Check the numbers on the four web pages (senis\_mms, metrolab\_hallinsight, un2015transforming, nasa\_smallsats) and keep the 9 Oct 2026 access date.
- ☐ Add to Zotero: Omelyanchik et al. 2023, *Meas. Sci. Technol.* 34, 107001 (Sobko's ref. [21]).
- ☐ Add to Zotero: a square-coil Helmholtz paper.
- ☐ Add to Zotero: a CubeSat Helmholtz-cage test source, if I-A needs one.
- ☐ **Batch 2 number assumed.** The request said "Batch <N>". Batch 1 was already in the index, so these six rows use 2. Change them if that's wrong.
- ☐ **Non-magnetic construction isn't unique to this project** (nogiec2022software). The Mu2e mapper works inside a 1 T field. It is built from aluminium, carbon fibre, plastic bearings and brass screws, and driven by non-magnetic ultrasonic motors. In II-E, frame the difference as low cost plus base-mounted servos >600 mm from the probe, not the non-magnetic build alone. The earlier flag still stands: Mu2e isn't commercial, so it doesn't source "commercial mappers put motors and steel next to the probe".
- ☐ **Tira-Thompson tested closed-loop digital servos** (tirathompson2008digital). Dynamixel AX-12s have position and load feedback; your servos are open-loop PWM. Use the report for the concepts (deflection grows with load; friction depends on direction), not as a like-for-like accuracy comparison (his  $0.36^\circ \sigma$  vs your  $3.4^\circ \rightarrow 0.5^\circ$  rms).
- ☐ **The ISO 9283 file is only the catalogue page** (iso1998manipulating). It has no definitions. Get the standard's full text for the accuracy and repeatability definitions (II-F, IV-A) and their clause numbers. It's under review, so recheck the edition before submission.
- ☐ **No source yet for the Jacobian-transpose step in III-H.** Williams gives only the velocity relation  $\dot{A}\mathbf{x} = \mathbf{B}\dot{\theta}$  ( $\mathbf{J} = \mathbf{A}^{-1}\mathbf{B}$ ). Mapping forces to joint torques through  $\mathbf{J}^T$  needs a robotics textbook or paper.
- ☐ **Which Nguyen paper?** (bentaleb2020improvement). Bentaleb cites Nguyen et al. 2015 (EKF + ANN for non-geometric error). Your outline cites "Nguyen 2022" in IV-J. Confirm which one you mean.

- ☐ **Joint-angle convention** (williams2016delta). Williams sets  $\theta = 0$  with the arm horizontal, and positive  $\theta$  moves the arm down. Check this against your firmware before quoting his equations or example angles.
- ☐ **Bentaleb errors not to repeat** (bentaleb2020improvement): Eq. (3.8) should subtract  $Lb_i^2$ , not  $Lb_i$ . The sign of (3.17) disagrees with (3.18). The radius names on p. 263 clash. The workspace study in the abstract isn't in the paper, and the 48.27 % gain in the Discussion matches no result.
- ☐ **Williams slips** (williams2016delta): p. 6 calls the platform frame the "base frame {P}". The prismatic trajectory is centred on  $-1$  m in the text (p. 37) but on  $-0.5$  m in the plots (p. 38). The date reads Oct. 2016 on the title page and Jan. 2016 in the citation line; cite Jan. 2016.
- ☐ **Tira-Thompson inconsistencies** (tirathompson2008digital): the Table 2 caption says milliradians, but the values are degrees. The text gives the rated 12 kgf·cm at 12 V, but Table 1 gives it at 7 V. The deflection models' sign convention and units aren't stated.
- ☐ **Mu2e arm spans have no units** (nogiec2022software, p. 2). They're probably inches; don't cite them.
- ☐ **Tira-Thompson reference is incomplete**. No institution or URL is printed. Find where you downloaded it (probably a CMU Tekkotsu page) and add the URL and access date.
- ☐ Verify nogiec2022software requirement and spec numbers (pp. 1-2 and 4: 0.01 %, 0.1 mrad,  $\pm 0.005$  mm, 200 G, 0.07 mm) against the PDF.
- ☐ Verify clavel1990device col. 3, lines 1-11 and col. 4, lines 37-44 (p. 7), and the filing and priority dates on p. 1.
- ☐ Verify williams2016delta equations on pp. 6, 7, 10-13 and 15, Eqs. (B.9)-(B.12) on pp. 45-46, and the FlexPicker table on p. 6. The worked examples on pp. 16 and 19 already reproduce numerically.
- ☐ Verify bentaleb2020improvement Eqs. (3.1)-(3.21) and (4.1) (pp. 264-268) and the gain figures in the text (pp. 268-270).
- ☐ Verify tirathompson2008digital Tables 1-4 (pp. 2, 3 and 5) and the deflection and mass models (pp. 6-7).
- ☐ Check the iso1998manipulating page details (edition, 60 pp., confirmed 2021, stage 90.60) and keep the 9 Oct 2026 access date.
- ☐ Add to Zotero: Clavel 1991, EPFL PhD thesis "Conception d'un robot parallèle rapide à 4 degrés de liberté" (Williams, fn. 1).
- ☐ Add to Zotero: Feher et al. 2018, "Mu2e field mapping system design," *IEEE Trans. Appl. Supercond.* 28(3), 9001305 (Nogiec ref. [3]).

- ☐ Add to Zotero: Orozco et al. 2018, Hall probe calibration for the Mu2e mapper, *IEEE Trans. Appl. Supercond.* 28(3), 9500504 (Nogiec ref. [7]) — for IV-H probe calibration.
- ☐ Add to Zotero: Lippitsch et al., Leica absolute laser tracker operation in magnetic fields (IWAA2018 paper; Nogiec ref. [9]) — for II-E external metrology.
- ☐ Add to Zotero: Bai et al. 2015, kinematic calibration of a Delta robot using distance measurements, *Proc. IMechE Part C* 230(3), 414–424 (Bentaleb ref. 4).
- ☐ Add to Zotero: Kamali et al. 2016, elasto-geometrical calibration of an industrial robot under external loads using a laser tracker, *ICRA*, 4320–4327 (Bentaleb ref. 11) — joint stiffness for II-D and III-H.
- ☐ Add to Zotero: Zhao et al. 2015, metrological evaluation and kinematic calibration of a medical parallel robot, *Int. J. Adv. Robotic Systems* 12(9), 1–13 (Bentaleb ref. 23).
- ☐ Add to Zotero, if it's the Nguyen you need: Nguyen, Zhou and Kang 2015, EKF + ANN robot calibration, *Neurocomputing* 151, 996–1005 (Bentaleb ref. 16).
- ☐ Add to Zotero: Williams, Albus and Bostelman 2004, "3D cable-based Cartesian metrology system," *J. Robotic Systems* 21(5), 237–257 (source of Williams' Appendix A).
- ☐ Add to Zotero: Lange and Hirzinger 1996, "Learning force control with position controlled robots," *ICRA* (Tira-Thompson ref. [4]).
- ☐ Add to Zotero: a robotics text or paper for the statics relation  $\tau = J^T f$ , for III-H.
- ☐ **IGRF-13 has expired** (alken2021international). It was valid only to 2025.0. IGRF-14 was released on 19 Nov. 2024 (DGRF 2020, IGRF 2025, secular variation 2025–2030). Use IGRF-14 (or WMM) for any Wellington field value in 2026, and keep Alken only for the model definition until you have the IGRF-14 paper.
- ☐ **The NOAA calculator file has no result** (noaa\_magcalc). It's the empty form, with the Declination tab open and N/W hemispheres. Rerun it on the Magnetic Field tab with your lab's coordinates (S, E), the scan date and the model, then save the PDF or CSV and record the model used.
- ☐ **Don't use Uscategui's "d = L" spacing for square coils** (uscategui2023highprecision, p. 5). My calculation with their Eq. (13) puts the uniform-field spacing at about 0.5445 L. Their own cage uses 0.5 m for 1.03 m sides (about 0.49 L), which contradicts the text. This matters for the Wilson square-coil comparison (IV-K, VI). Confirm the spacing with a primary source.
- ☐ **Uscategui mixes circular and square coil formulas.** Eq. (13) is for square coils, but Eqs. (14)–(15) and the current formula (28)–(30) are for circular coils. Only Eq. (13) applies to Wilson's square cage.

- ☐ **Uscategui's Table 7 uniformity is better than ideal theory** (p. 19). They report  $\leq 0.12\%$  over  $\pm 13$  cm, but ideal thin loops at their geometry give about  $1.25\%$  on axis at 12.5 cm (about  $0.27\%$  even at optimum spacing). Treat the figure with caution, or don't use it as a benchmark.
- ☐ **Uscategui's ">99.89 % accuracy" is control-loop tracking at the centre point** (Tables 5–6). It's measured by the same magnetometer that closes the PID loop, so don't cite it as cage accuracy or uniformity.
- ☐ **Uscategui slips not to repeat:** the hard- and soft-iron descriptions are swapped on p. 7 (the maths is right);  $\epsilon\%$  is called "permeability" (p. 6); Eq. (8) should have  $V$ , not  $\nabla V$ , on the left; Eq. (20) has  $a_{32}$  where  $a_{23}$  is meant; Eq. (3)'s brackets don't balance; p. 19 says "Table 6" but means Table 7; Table 5 has "1,5293.10" for 15 293.10; Table 3 and Fig. 14 give different Y and Z currents; and the turns in Table 2 aren't integers.
- ☐ **Pollák & Goryl's figures aren't ISO 9283 RP\_I** (pollak2023simulation). Their Eq. (5) is  $\pm 3S$  per axis with no  $\Gamma$  term, Z was dropped, and they ran at 50 % load and 50 % speed. Don't compare their 0.03–0.09 mm directly with your  $\sim 2$  mm RP\_I without saying so, and state which metric your report uses (IV-D, IV-J).
- ☐ **Pollák & Goryl give the ISO 9283 AP and RP\_I formulas** (Eqs. (1)–(4), p. 5). That's a citable secondary source for the definitions in II-F, but it doesn't replace getting the standard's full text (the earlier task still stands). Their Eq. (4) is missing the "+" signs inside  $I_j$ .
- ☐ **Cervettini's torque numbers don't match Table I** (cervettini2021development, p. 1348).  $0.248 \text{ A}\cdot\text{m}^2$  at 1.705 G gives  $4.2 \times 10^{-5} \text{ N}\cdot\text{m}$ , not the  $6.5 \times 10^{-5} \text{ N}\cdot\text{m}$  stated, and at the unaugmented field it gives about  $1.5 \times 10^{-5} \text{ N}\cdot\text{m}$ , not  $\approx 10^{-6} \text{ N}\cdot\text{m}$ . Don't quote these torques.
- ☐ **CubeSat framing decision.** cervettini2021development and uscategui2023highprecision are the CubeSat Helmholtz-cage sources the earlier "Add to Zotero" task asked for. Using them in I-A still depends on whether the report says Wilson's cage is for CubeSat testing (the same condition as nasa\_smallsats). Tick the earlier task yourself if you're happy with these two.
- ☐ **Earth's-field values now conflict across batches.** Sobko gives about  $10 \mu\text{T}$  (Batch 1, already flagged as wrong). Cervettini's lab was about  $62 \mu\text{T}$  (California), Uscategui's site  $54.8 \mu\text{T}$  (Beijing), and IGRF Fig. 1 puts NZ in roughly the  $55\text{--}60 \mu\text{T}$  band. Quote only a value computed for Wellington (see the NOAA task).
- ☐ **No source yet for daily (external) geomagnetic variation.** IGRF supports "secular variation  $\leq \sim 100 \text{ nT/yr}$ , so  $\sim 1 \mu\text{T/h}$  drift isn't geomagnetic" (IV-G), but IGRF leaves out daily variation, so that part of the argument needs its own source.

- ☐ Verify cervettini2021development Eqs. (1)–(5) and the J matrix (pp. 1347–1348), Table I (p. 1346), and the torque and field figures on p. 1348.
- ☐ Verify pollak2023simulation Eqs. (1)–(5) (pp. 5 and 12), Tables 1–2 (pp. 4 and 7) and Table 5 (p. 12). Tables 3–4 (pp. 10–11) already reproduce Table 5.
- ☐ Verify uscategui2023highprecision Eqs. (1)–(30) (pp. 3–7 and 13–14), the A and b calibration values (p. 15), Tables 2–3 (pp. 9–10), Tables 5–7 (pp. 17 and 19), and the Fig. 8–10 marker values (pp. 11–12). Check the exponent placement in Eq. (13) in particular.
- ☐ Verify alken2021international Eqs. (1)–(6) (pp. 2, 14 and 21), Table 1 (p. 2), the 2020–2025 rows of Table 4 (p. 21), and the Fig. 2 dF/dt range (p. 19).
- ☐ Check the noaa\_magcalc page details (model list and ranges, the 30-arc-minute statement) and keep the 9 Oct 2026 access date.
- ☐ Add to Zotero: R. C. da Silva, I. S. K. Ishioka, C. Cappelletti, S. Battistini and R. A. Borges, "Helmholtz cage design and validation for nanosatellites HWIL testing," *IEEE Trans. Aerosp. Electron. Syst.*, 55(6), 3050–3061, 2019 (Cervettini ref. [5], Uscategui ref. [32]). For I-A and II-A.
- ☐ Add to Zotero: F. Piergentili, G. P. Candini and M. Zannoni, "Design, manufacturing, and test of a real-time, three-axis magnetic field simulator," *IEEE Trans. Aerosp. Electron. Syst.*, 47(2), 1369–1379, 2011 (Cervettini ref. [15], Uscategui ref. [33]). For I-A field intensity and uniformity.
- ☐ Add to Zotero: V. Renaudin, M. H. Afzal and G. Lachapelle, "Complete triaxis magnetometer calibration in the magnetic domain," *J. Sensors*, 2010, 967245 (Uscategui ref. [29]). For IV-K and IV-H hard- and soft-iron calibration.
- ☐ Add to Zotero: S. Wang, G. Shi and M. Zhang, "Localization error of magnetic dipole by measuring remote magnetic field and field-gradient tensor," *ICEMI 2019*, pp. 779–786 (Uscategui ref. [26]). For IV-F dipole localisation.
- ☐ Add to Zotero: S. Wang, M. Zhang, N. Zhang and Q. Guo, "Calculation and correction of magnetic object positioning error caused by magnetic field gradient tensor measurement," *J. Syst. Eng. Electron.*, 29, 456–461, 2018 (Uscategui ref. [27]). For IV-F and IV-E.
- ☐ Add to Zotero: L. Yue *et al.*, "A comprehensive calibration method for non-orthogonal error and scale factor error of triaxial Helmholtz coil," *Rev. Sci. Instrum.*, 92, 085105, 2021 (Uscategui ref. [28]). For IV-K and VI.
- ☐ Add to Zotero: M. Slamani, A. Nubiola and I. Bonev, "Assessment of the positioning performance of an industrial robot," *Ind. Robot*, 39(1), 57–68, 2012 (Pollák ref. [13]). For II-F and IV-A, ISO 9283 test practice.

- ☐ Add to Zotero: M. Vocetka *et al.*, "Influence of drift on robot repeatability and its compensation," *Appl. Sci.*, 11, 10813, 2021 (Pollák ref. [15]). For IV-D drift.
- ☐ Add to Zotero: J. Józwik, D. Ostrowski, P. Jarosz and D. Mika, "Industrial robot repeatability testing with high speed camera Phantom V2511," *Adv. Sci. Technol. Res. J.*, 10, 86–96, 2016 (Pollák ref. [14]). For IV-D comparison.
- ☐ Add to Zotero: the IGRF-14 definition paper (*Earth Planets Space* special issue "International Geomagnetic Reference Field – The Fourteenth Generation"); its companion is C. D. Beggan *et al.*, *Earth Planets Space* 78, 126, 2026, DOI 10.1186/s40623-026-02382-2. For IV-G.
- ☐ **Rod-end housing may be steel too** (nzminiaturebearings2026female). The listing says the ball is GCR15 bearing steel and the housing aluminium, but SI.T/K housings are often steel. The outline (IV-K) says only the balls are steel. Magnet-test a rod end and correct IV-K if needed.
- ☐ **"Electrically non-conductive" vs carbon fibre and aluminium** (tsekos2007magnetic, p. 358). Tsekos wrongly lists carbon fibre as non-conductive; it conducts, as does an aluminium top plate. If Table I requires non-conductive parts, reword to "non-ferromagnetic" or justify it (for a static map, conductivity only matters for eddy currents while moving). Mu2e (nogiec2022software) also uses carbon fibre and aluminium.
- ☐ **MRI terminology differs between Batch 4 sources.** tsekos2007magnetic uses "MR safe / MR compatible" (p. 357); farooq2023decade uses ASTM F2503 "MR Safe / Conditional / Unsafe" and says "MRI compatible" isn't an FDA term (p. 862). Pick one set for II-B.
- ☐ **Omelyanchik backs the narrowed cost wording** (omelyanchik2023doityourself, p. 2). Its ">€25,000" is for 3D mapping systems, which fits the Senis 3D mapper (US\$143,180) but not the US\$17,950 HallinSight camera. Follow-up to the earlier ">€25,000" flag. Its refs [9] GMW MMS-1A-RS and [10] SENIS are the senis\_mms product, so no new Zotero items.
- ☐ **Omelyanchik's "<€200" and 0.012 mm aren't measured results.** The €200 is a projection (Python + Arduino + cheap sensor); the built system used a Lake Shore 475 and LabVIEW. The 0.012 mm is the printer maker's figure. Say so if you set them beside your ISO 9283 results (II-E, IV-D).
- ☐ **Omelyanchik isn't an Earth's-field source** (p. 7). It calls ~0.3 G "the order" of the ambient field. Adds to the earlier Earth's-field conflict flag; still quote only a value computed for Wellington.

- ☐ **Omelyanchik only partly helps the "motors and steel" claim** (p. 7). It says printer mappers need their steel replaced to measure below  $\sim 1$  G, but that's a DIY printer, not a commercial mapper. The earlier flag stands.
- ☐ **Batch 1 Omelyanchik task can be ticked.** omelyanchik2023doityourself is now in the Index (Batch 4); tick the earlier "Add to Zotero: Omelyanchik et al. 2023" task yourself if you're happy.
- ☐ **Omelyanchik slips not to repeat:** p. 4 says  $\sigma$  "increases to its maximum" for  $t_d \leq 5$  s, but Fig. 4(b) shows  $\sigma$  largest at the shortest delay (quote the figure); "N = 10 s" should be N = 10 readings (p. 4); "lower resolution" means finer (p. 7); Fig. 6 panel heights 0/4/8 mm vs caption 1.9/5.9/9.9 mm (1.9 mm offset); accuracy 0.012 mm (p. 2) vs " $>0.012$  mm" (p. 4).
- ☐ **Farooq & Ko slips not to repeat:** abstract 2009–2021 vs search 2008–2021 (pp. 862–863); Table IV gives ref. [160] to both Jiang and Giannakou and puts "20  $\mu\text{m}$ " in a mm column (p. 873); Table V gives Stormram 4 0.7 mm vs  $1.29 \pm 0.59$  mm in the text (pp. 873–874). Its "accuracy" mixes targeting and positioning; label it if compared with your 2.3 mm rms (IV-D).
- ☐ **Tsekos slips:** "0.3 open scanner" lacks the unit T (p. 366); ref. 98's pages are misprinted "2886–71" (p. 386); the web page date (15 Aug. 2007) differs from the PDF's online date (17 Apr. 2007). The PDF and web page are one article: one Zotero item with both attachments.
- ☐ **Wellington field cross-check** (alken2021international, Batch 4 check). IGRF-14 via the ppigrf Python package gives  $F \approx 55.6 \mu\text{T}$ ,  $I \approx -66.5^\circ$ ,  $D \approx +23.3^\circ$  for  $41.29^\circ$  S,  $174.78^\circ$  E on 1 Oct. 2026 ( $F \approx 55.9 \mu\text{T}$  at 2020.0, about  $-36$  nT/yr). Use it only to check the NOAA rerun; don't cite it. Eq. (6) with Table 2's 2020.0 dipole terms (p. 8) gives  $7.71 \times 10^{22} \text{ A}\cdot\text{m}^2$ , matching Fig. 5.
- ☐ Verify omelyanchik2023doityourself  $z_{\text{active}} = z + 1.9$  mm (p. 2), the arrow-end formula and Figs. 4(b) and 5 (p. 5), and the cost statements (pp. 2 and 7) against the PDF.
- ☐ Verify tsekos2007magnetic pp. 358 (materials, small metal parts), 360 (remote-actuation backlash), 362 (four-condition test) and 363–364 (accuracy values) against the PDF.
- ☐ Verify farooq2023decade actuator percentages (p. 878), ASTM F2503 terms (p. 862), refs [144], [183] and [193] (pp. 872–874), and any Table IV–VI values you quote (pp. 873–875).
- ☐ Check nzminiaturebearings2026female against the manufacturer drawing or live page: tilt angle  $13^\circ$  (no unit on the page), dimensions, housing material; keep the 9 Oct 2026 access date.

- ☐ Add to Zotero: K. Erglis, J. Cimurs and G. Kitenbergs, "Low-cost solution for magnetic field mapping device," 22nd Int. Conf. Electronics (IEEE), 2018, pp. 1–5 (Omelyanchik ref. [8]). For II-A.
- ☐ Add to Zotero: A. Vavoulas et al., "Using a modified 3D-printer for mapping the magnetic field of RF coils designed for fetal and neonatal imaging," *J. Magn. Reson.* 269, 146–151, 2016 (Omelyanchik ref. [13]). For II-A.
- ☐ Add to Zotero: H. Han et al., "Open source 3D multipurpose measurement system with submillimetre fidelity and first application in magnetic resonance," *Sci. Rep.* 7, 2017 (Omelyanchik ref. [14]). For II-A.
- ☐ Add to Zotero: J. F. Schenck, "The role of magnetic susceptibility in magnetic resonance imaging," *Med. Phys.* 23, 815–850, 1996 (Tsekos ref. 88). For III-C material magnetic properties.
- ☐ Add to Zotero: E. Christoforou and N. V. Tsekos, "Robotic manipulators with remotely-actuated joints: implementation using drive shafts and u-joints," ICRA 2006 (Tsekos ref. 98; check pages). For III-F.
- ☐ Add to Zotero: K. Chinzei et al., MR compatible surgical assist robot, MICCAI 2000, pp. 921–930, and Chinzei et al., MR compatibility of mechatronic devices: design criteria, MICCAI 1999, pp. 1020–1031 (Tsekos refs. 8, 86). For II-B.
- ☐ Add to Zotero: Y. Koseki et al., "Precise evaluation of positioning repeatability of MR-compatible manipulator inside MRI," MICCAI 2004, pp. 192–199 (Tsekos ref. 99). For IV-D and II-E.
- ☐ Add to Zotero: S. Elayaperumal et al., "A passive parallel master–slave mechanism for magnetic resonance imaging-guided interventions," *J. Med. Devices* 9(1), 2015 (Farooq ref. [144]). For II-B and II-C.
- ☐ Add to Zotero: X. Wang, S. S. Cheng and J. P. Desai, "Design, analysis, and evaluation of a remotely actuated MRI-compatible neurosurgical robot," *IEEE Robot. Autom. Lett.* 3(3), 2144–2151, 2018 (Farooq ref. [183]). For II-B and IV-D.
- ☐ Add to Zotero: K. D. Price et al., "Design and validation of an MR-conditional robot for transcranial focused ultrasound surgery in infants," *Med. Phys.* 43(9), 4983–4995, 2016 (Farooq ref. [193]). For III-F.
- ☐ Add to Zotero: H. Elhawary et al., "The case for MR-compatible robotics: a review of the state of the art," *Int. J. Med. Robot.* 4(2), 105–113, 2008 (Farooq ref. [14]). For II-B.
- ☐ Add to Zotero: Q. Xiao et al., "MR-conditional actuations: a review," *Ann. Biomed. Eng.* 48(12), 2707–2733, 2020 (Farooq ref. [25]). For II-B.
- ☐ Add to Zotero: ISO 12240-4 (rod ends) and the SI6T/K manufacturer datasheet. For III-B and III-C.

- ☐ Add to Zotero: GNS Eyrewell (EYR) observatory data for your mapping days (alken2021international Table 3, p. 17). A candidate source for the earlier daily-variation flag (IV-G).
- ☐ **Batch 5's web page and PDF are one article** (yang2020analytical). The ScienceDirect page S0307904X19306122 is Yang et al. 2020, so it has one Index row and one note. Make one Zotero item with both attachments; use the PDF for page numbers and the web page for the URL and the 9 Oct 2026 access date.
- ☐ **Batch 5 cite keys are provisional** (yang2020analytical, futaba2010s3306mg). Swap them for your .bib keys; follows the earlier provisional-keys task.
- ☐ **Yang Eqs. (29), (37), (39) look wrong** (yang2020analytical, pp. 331, 333). The rotational terms  $8EA_y b/(Lg)$ ,  $16EA_z b/(Lg)$  and  $12EA_z b/(Lg)$  give N/m, not N·m/rad. Eq. (16)'s  $4EI_y b_1/(Lg_2)$  implies  $I_y$ ,  $I_z$ , not  $A_y$ ,  $A_z$ . Don't copy them.
- ☐ **Yang Eq. (19) notation** (p. 330).  $K_{112}$  is marked "sym.", but a beam's off-diagonal block isn't symmetric (the  $6EI$  couplings change sign). The rotational entries check out; it doesn't affect the report.
- ☐ **Yang Table 1 units unclear** (p. 335). Columns labelled "(mm)" with values around  $10^{-7}$  look like compliance, but the text says stiffness diagonal; it also uses "R" where Eq. (41) uses "r".
- ☐ **Yang's conclusion understates the error** (p. 348 says  $< 8\%$ ). Table 4 (p. 341) shows 8.87 % and Section 3.5 says  $< 9\%$ . Quote 8.87 % or "under 9 %".
- ☐ **Yang Table 5 likely sign typo** (p. 342). Config 1, scenario 3 has  $\Delta l_z = +0.010$  while every other entry is negative; the 39.84 % error only fits the magnitude. Config 1, scenario 2  $\Delta l_x$  error recomputes to 4.19 %, not 4.23 %. The web version has the same values.
- ☐ **Yang slips not to repeat:** Eq. (8) ends with  $w^i$  for  $w_i$  (p. 327); nomenclature " $\delta_B i$ " and " $UB i$ " (p. 323); Figs. 17–18 captions say "(a) m11 and m31" for (c) (p. 345); Fig. 13 legend "regualr" (p. 342).
- ☐ **Two different "J"s across batches** (yang2020analytical vs williams2016delta). Yang's  $J_i$  is a  $6 \times a_i$  matrix of unit constraint screws; Williams's  $J = A^{-1}B$  is the delta's velocity Jacobian. Don't cite Yang for the kinematic Jacobian in III-E, and don't mix the two in III-H.
- ☐ **Yang assumes rigid actuators and no gravity** (pp. 324–325), which conflicts with III-H. Your correction is for load-induced deflection in a servo-driven delta. Cite Yang only for the general idea (compliance mapped through Jacobian-type matrices, Eqs. (45)–(47)); for servo or joint stiffness use tirathompson2008digital's torque-to-deflection model plus Gosselin 1990 (below).

- ☐ **The III-H Jacobian-transpose gap is still open.** Yang's  $\delta_i = J_i^T \Delta$  is for constraint screws with rigid actuators, so it doesn't source  $\tau = J^T f$  for your servos. Gosselin 1990 (below) is the candidate for the earlier "robotics text or paper for  $\tau = J^T f$ " task.
- ☐ **Yang's results are FEA-only on a large steel machine** (2PRU-UPR, 60 mm links,  $E = 210$  GPa). There's no experiment, so don't present its percentages as typical for low-cost robots (II-D, IV-J).
- ☐ **Optional: Yang's S-joint model for the rod ends** (Eqs. (40)–(41), p. 334;  $r_s = 0.635$  r). It could estimate the steel-ball rod ends' translational compliance, but  $\zeta$  was fitted only for solid steel spheres. Ties in with the earlier "rod-end housing may be steel" flag (nzminiaturebearings2026female).
- ☐ **Futaba datasheet has no source details** (futaba2010s3306mg). Add the download URL and access date before citing.
- ☐ **Your servo measurements aren't in the datasheet** (futaba2010s3306mg). The 800–2200  $\mu$ s travel (working  $\sim 830$ – $2170$   $\mu$ s), 1520  $\mu$ s neutral and  $\sim 4^\circ$  output play are your measurements; cite them as such.
- ☐ **No servo deadband or backlash source yet for IV-B.** The Futaba datasheet gives no deadband, backlash or hysteresis figures.
- ☐ **Futaba ratings are at 6.0 V** (III-D). A  $4 \times$  AA pack sags under load, so real torque and speed may be below the rating; state "at 6.0 V". (Tira-Thompson's ratings also depend on voltage; see the earlier flag.)
- ☐ **Futaba torque units** (futaba2010s3306mg). Give  $\approx 2.35$  N·m in the IEEE report, with 24.0 kg·cm in brackets if wanted.
- ☐ **Futaba says only "metal gear"** (futaba2010s3306mg). It doesn't mention the motor magnets; cite it only for metal gears when justifying servo placement more than 600 mm above the probe.
- ☐ Verify yang2020analytical Eqs. (29), (37), (39) (pp. 331 and 333) against the PDF.
- ☐ Verify yang2020analytical Eqs. (45)–(47) (p. 335), the main result cited in III-H.
- ☐ Verify yang2020analytical Table 1 (p. 335), Tables 2–3 (p. 340), Table 4 (p. 341) and Table 5 (p. 342).
- ☐ Verify the futaba2010s3306mg spec box (p. 1).
- ☐ Add to Zotero: C. Gosselin, "Stiffness mapping for parallel manipulators," *IEEE Trans. Robot. Autom.*, 6, 377–382, 1990 (Yang ref. [4]). Actuator stiffness through the Jacobian; for III-H.

- ☐ Add to Zotero: D. Deblaise, X. Hernot and P. Maurine, "A systematic analytical method for PKM stiffness matrix calculation," ICRA 2006, pp. 4213–4219, doi 10.1109/ROBOT.2006.1642350 (Yang ref. [28]). Delta stiffness model; for II-C and II-D.
- ☐ Add to Zotero: W. Yoon, T. Suehiro, Y. Tsumaki and M. Uchiyama, "Stiffness analysis and design of a compact modified delta parallel mechanism," *Robotica*, 22, 463–475, 2004 (Yang ref. [29]). For II-C and II-D.
- ☐ Add to Zotero: S. Yan, S. Ong and A. Nee, "Stiffness analysis of parallelogram-type parallel manipulators using a strain energy method," *Robot. Comput.-Integr. Manuf.*, 37, 13–22, 2016 (Yang ref. [26]). Delta-type parallelograms; for II-D.
- ☐ Add to Zotero: A. Klimchik, D. Chablat and A. Pashkevich, "Stiffness modeling for perfect and non-perfect parallel manipulators under internal and external loadings," *Mech. Mach. Theory*, 79, 1–28, 2014 (Yang ref. [19]). Virtual-joint and joint-stiffness models; for III-H.
- ☐ Add to Zotero: A. Pashkevich, A. Klimchik and D. Chablat, "Enhanced stiffness modeling of manipulators with passive joints," *Mech. Mach. Theory*, 46, 662–679, 2011 (Yang ref. [20]). For III-H.
- ☐ Add to Zotero: A. Klimchik, A. Pashkevich and D. Chablat, "Fundamentals of manipulator stiffness modeling using matrix structural analysis," *Mech. Mach. Theory*, 133, 365–394, 2019 (Yang ref. [34]). Review; for II-D.
- ☐ Add to Zotero (lower priority): C. Yang, Q. Li, Q. Chen and L. Xu, "Elastostatic stiffness modeling of overconstrained parallel manipulators," *Mech. Mach. Theory*, 122, 58–74, 2018 (Yang ref. [27]); and H. Liu et al., "Stiffness modeling of parallel mechanisms at limb and joint/link levels," *IEEE Trans. Robot.*, 33, 734–741, 2017 (Yang ref. [33]).
- ☐ Add to Zotero (optional): C. Yang et al., "Review of the performance optimization of parallel manipulators," *Mech. Mach. Theory*, 170, 104725, 2022 (cites Yang 2020). Possible II-D review; N. Ye and B. Hu, *Mech. Mach. Theory*, 172, 104821, 2022 is background only.